



Solvable quantum circuits from spacetime lattices

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arXiv:2512.15871

Motivation



- Thermalization, chaos, scrambling,...

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- Exactly solvable models of many-body quantum chaos: always have special properties
 - What are those? General theory? How can dual unitarity be generalized?

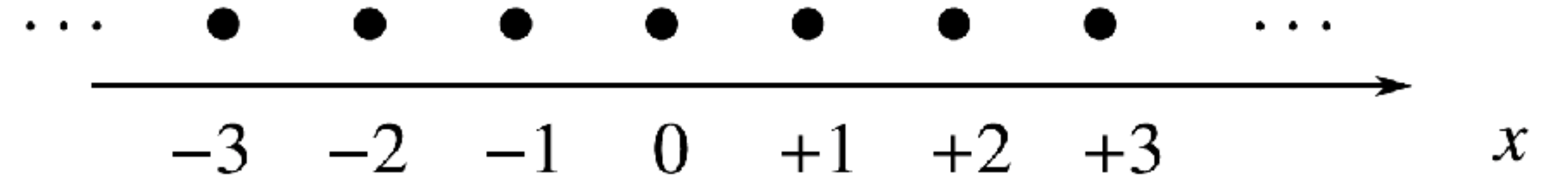
Unitary Circuits



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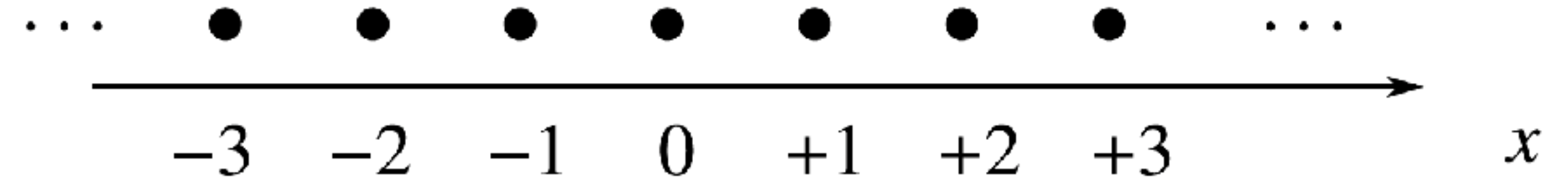
- Chain of qudits (dimension q)



Unitary Circuits



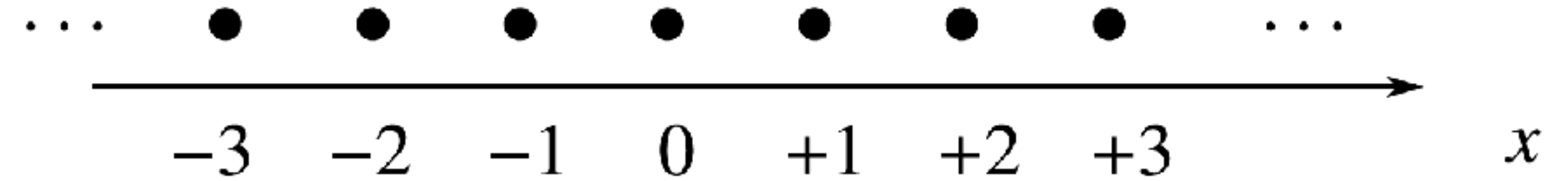
- Chain of qudits (dimension q)
- Two-site gates act on local Hilbert space



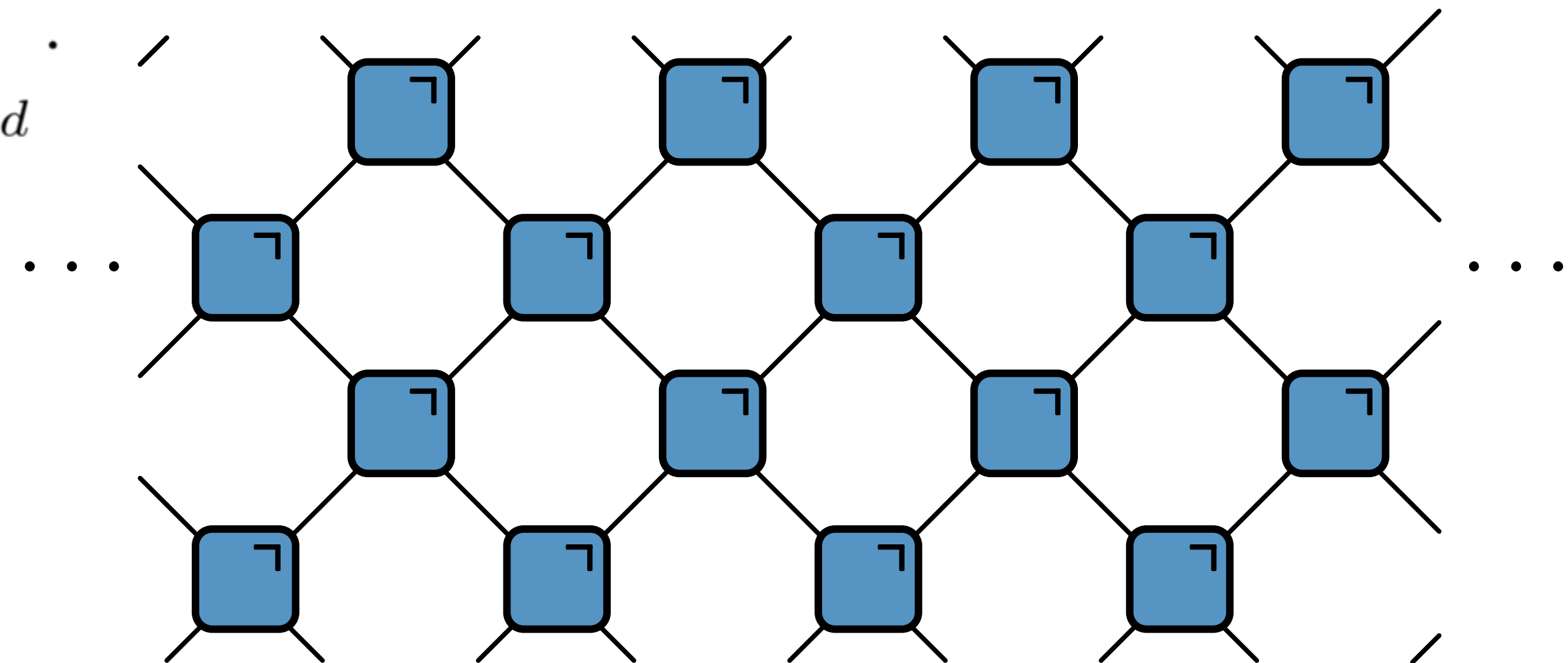
$$U_{ab,cd} = \text{blue square with legs } a, b, c, d, \text{ and a small square in the top-right corner}, \quad U_{ab,cd}^\dagger = \text{red square with legs } a, b, c, d, \text{ and a small square in the top-right corner}.$$

Unitary Circuits

- Chain of qudits (dimension q)
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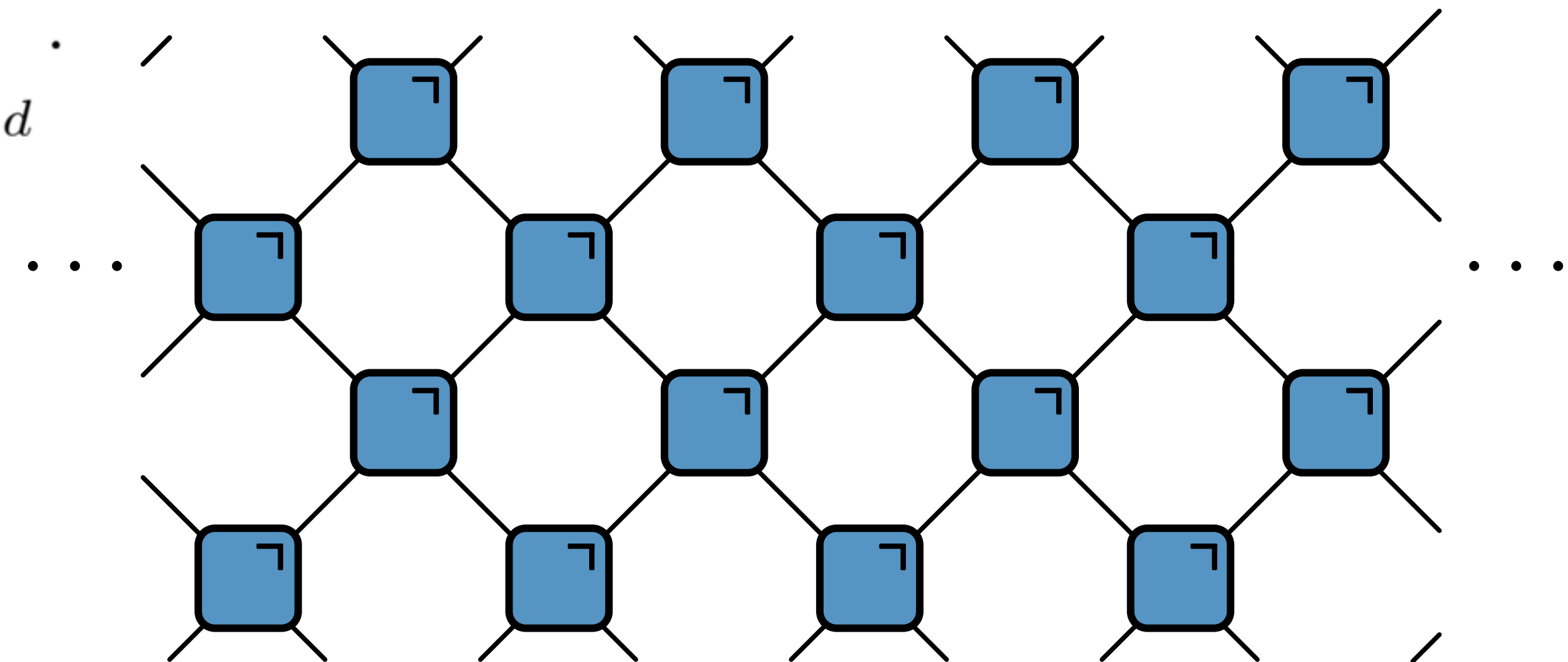
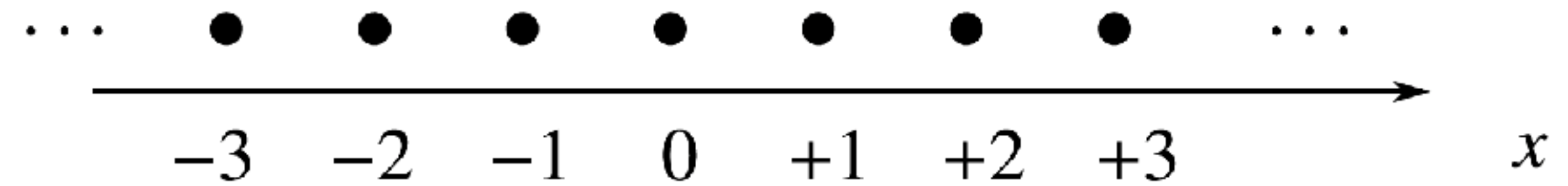


Unitary Circuits

- Chain of qudits (dimension q)
- Two-site gates act on local Hilbert space

$$U_{ab,cd} = \begin{array}{c} a \quad b \\ \diagdown \quad \diagup \\ \square \\ \diagup \quad \diagdown \\ c \quad d \end{array}, \quad U_{ab,cd}^\dagger = \begin{array}{c} a \quad b \\ \diagdown \quad \diagup \\ \square \\ \diagup \quad \diagdown \\ c \quad d \end{array}.$$

- Local and unitary dynamics



Entanglement Membrane Theory



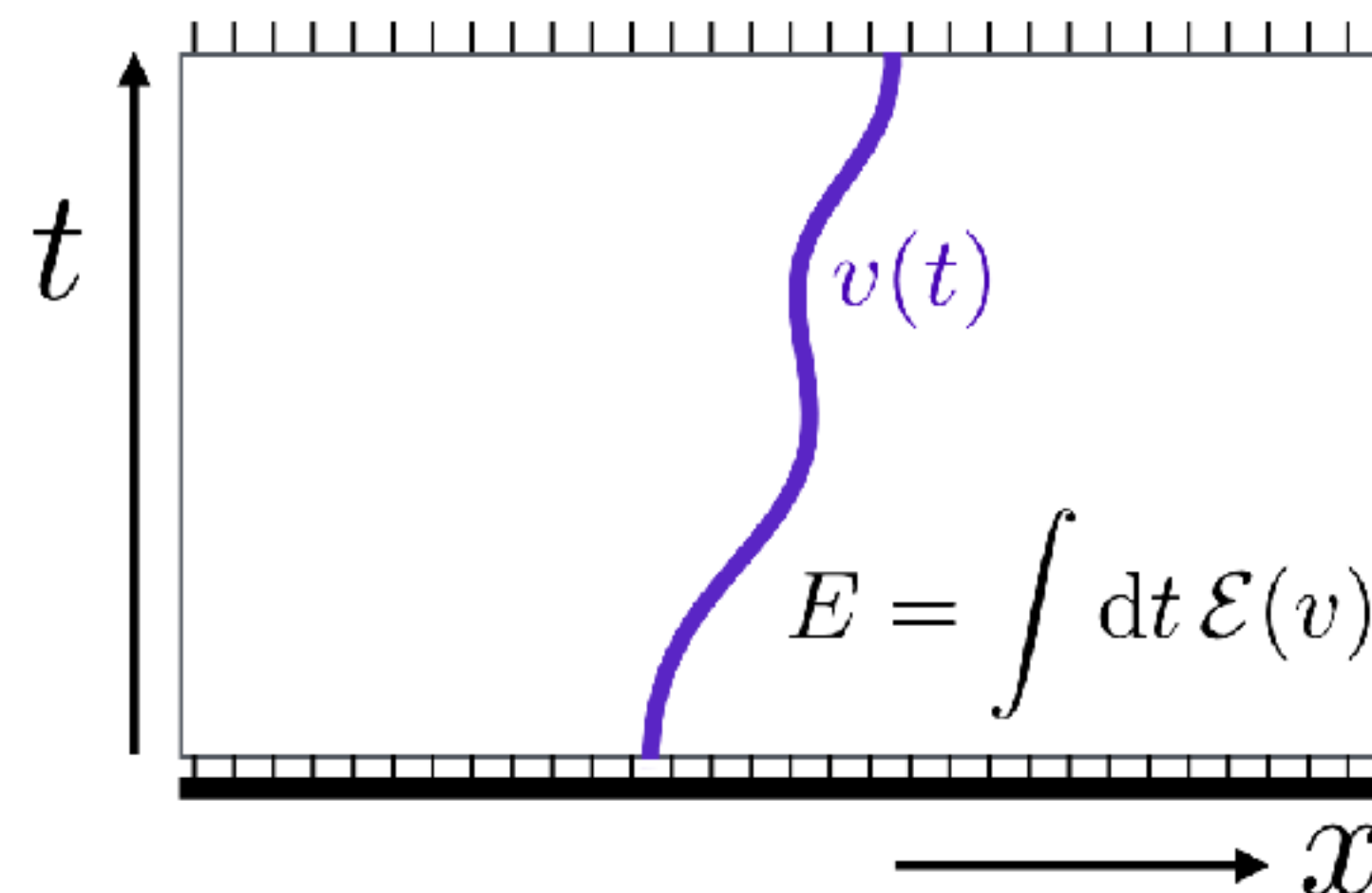
- Effective hydrodynamic theory of entanglement dynamics in chaotic systems at large wavelengths

Nahum et al. (2017),
Jonay, Huse, Nahum
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Entanglement Membrane Theory

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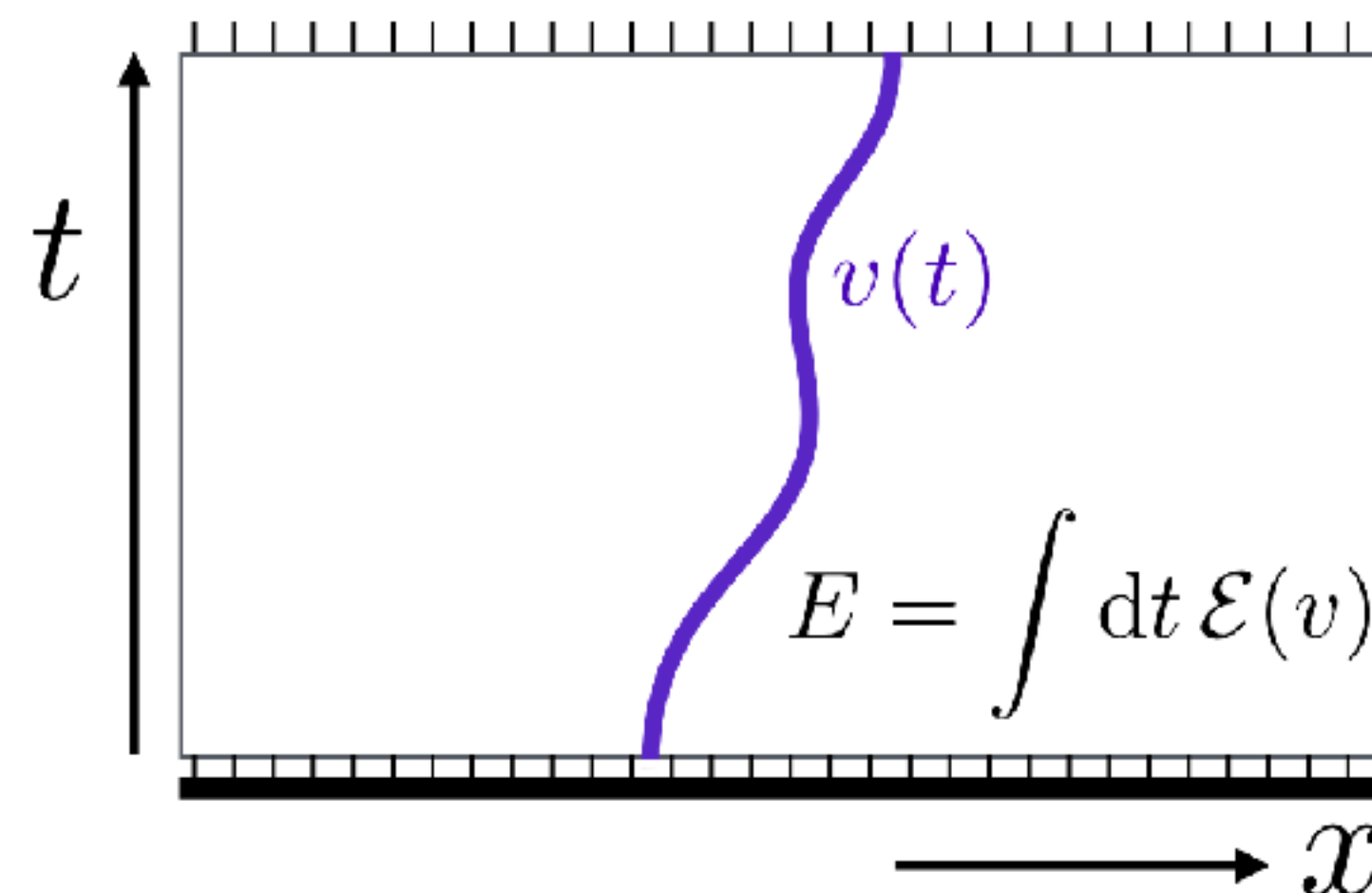


Entanglement Membrane Theory

- Effective hydrodynamic theory of entanglement dynamics in chaotic systems at large wavelengths
- Minimal configuration of effective membrane
- Ryu-Takayanagi formula in holography

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Ryu, Takayanagi (2006)



Entanglement Membrane Theory



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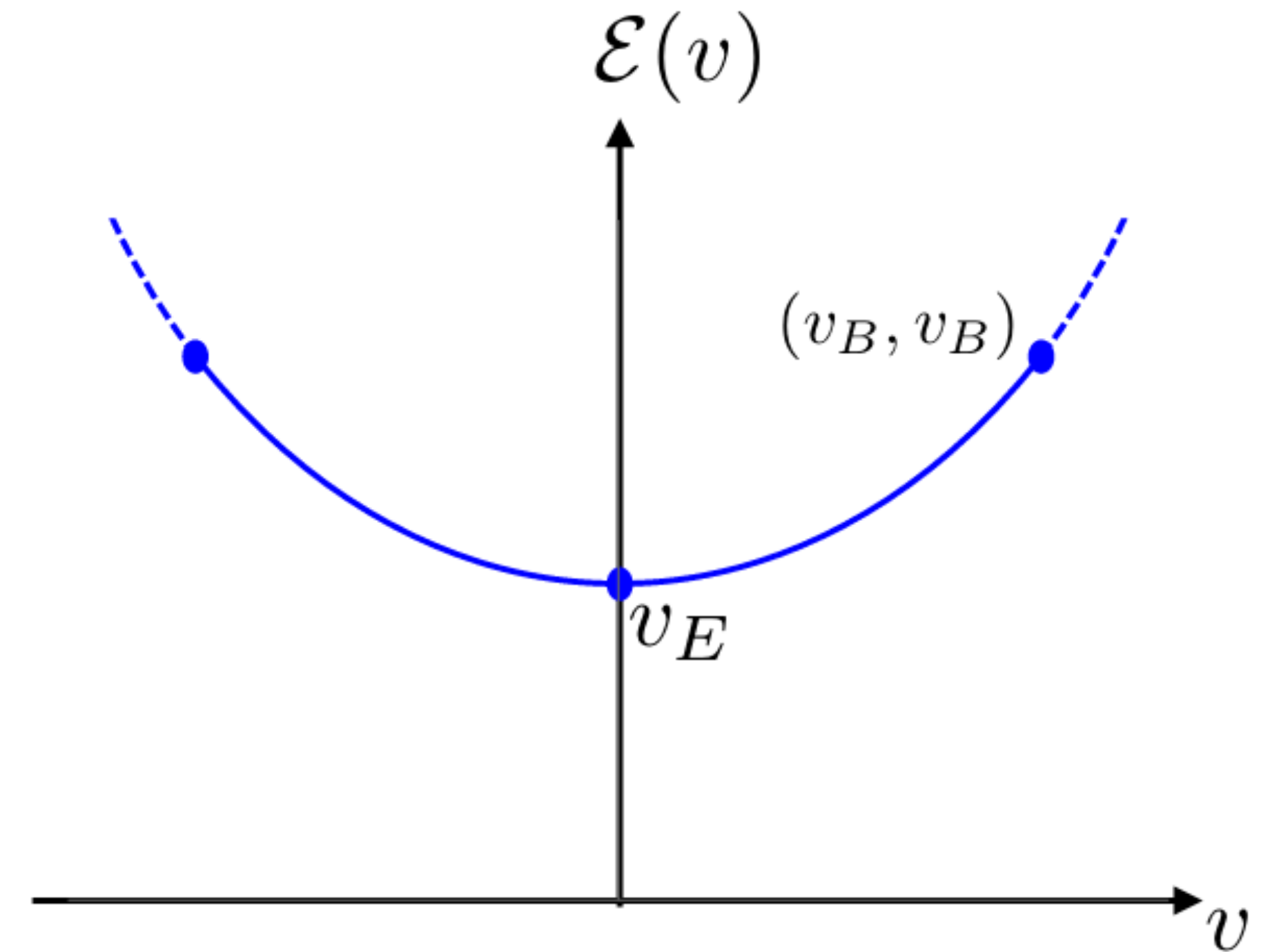
$$\frac{\partial S(x, t)}{\partial t} = s_{\text{eq}} \Gamma \left(\frac{\partial S(x, t)}{\partial x} \right)$$

- Legendre transformation yields minimal cut picture

$$\mathcal{E}(v) = \max_s \left(\Gamma(s) + \frac{vs}{s_{\text{eq}}} \right)$$

Entanglement Line Tension

- Microscopic information: entanglement line tension

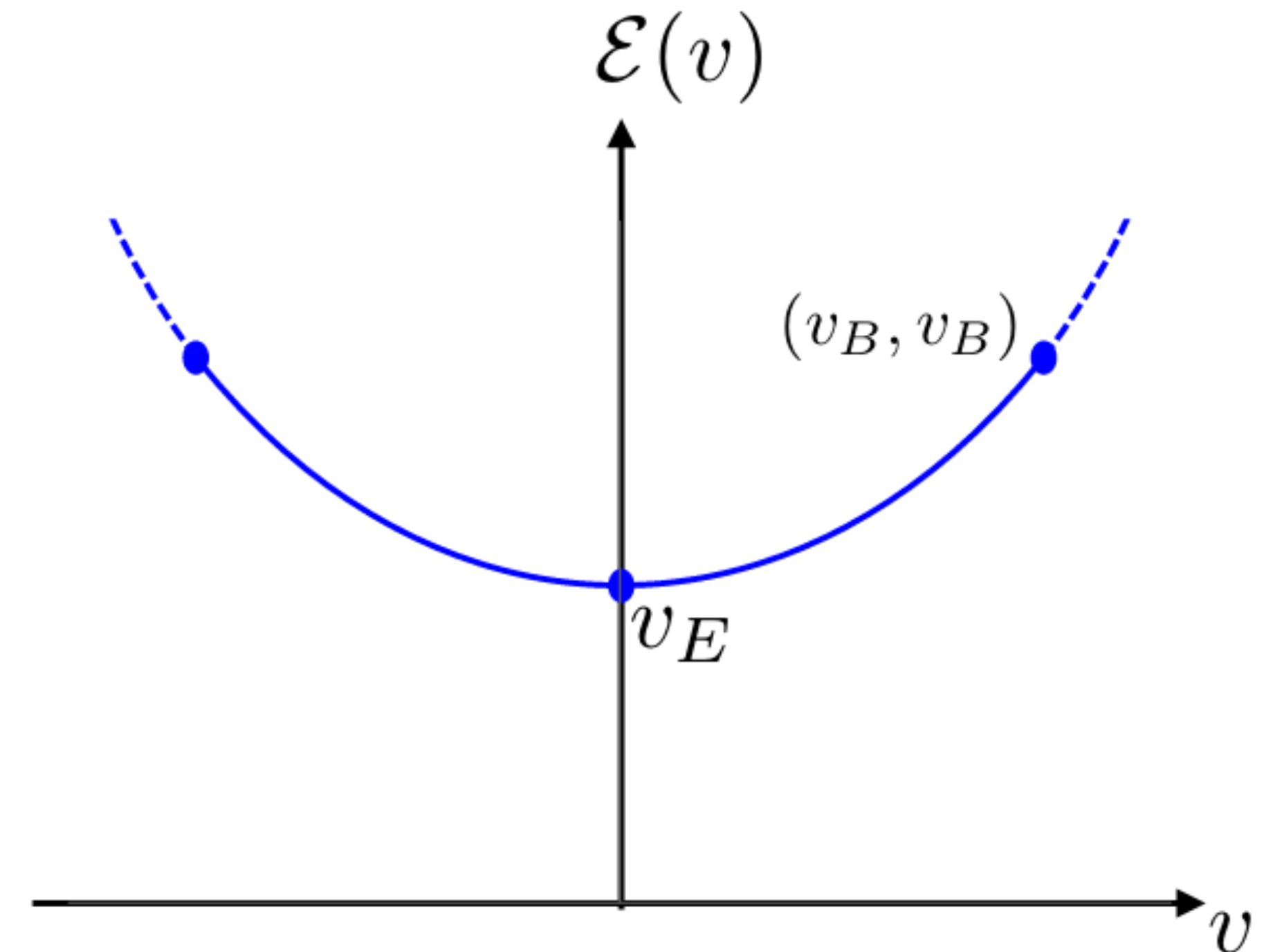


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Entanglement Line Tension

- Microscopic information: entanglement line tension
- Interpretation: information that passes through membrane

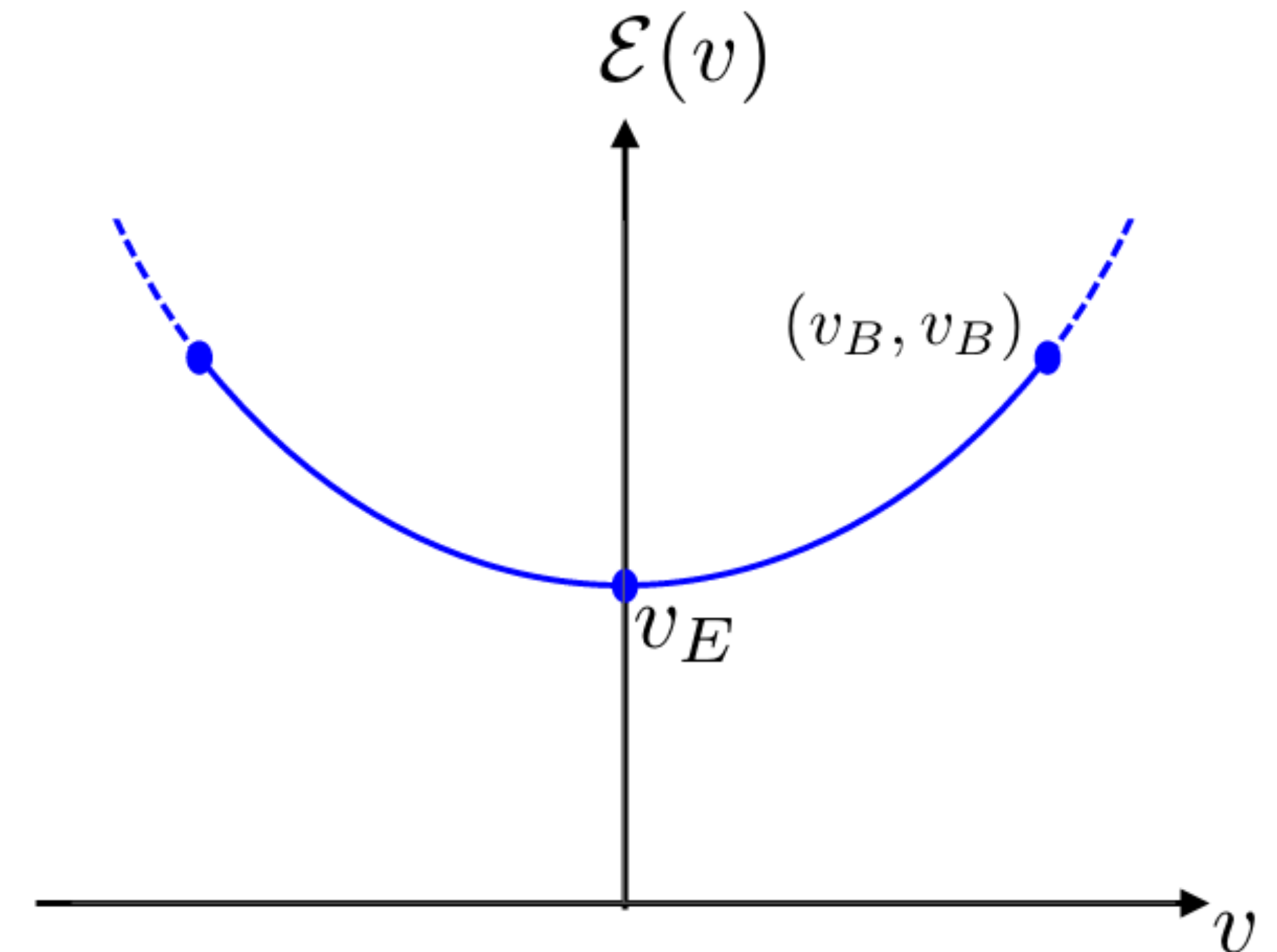
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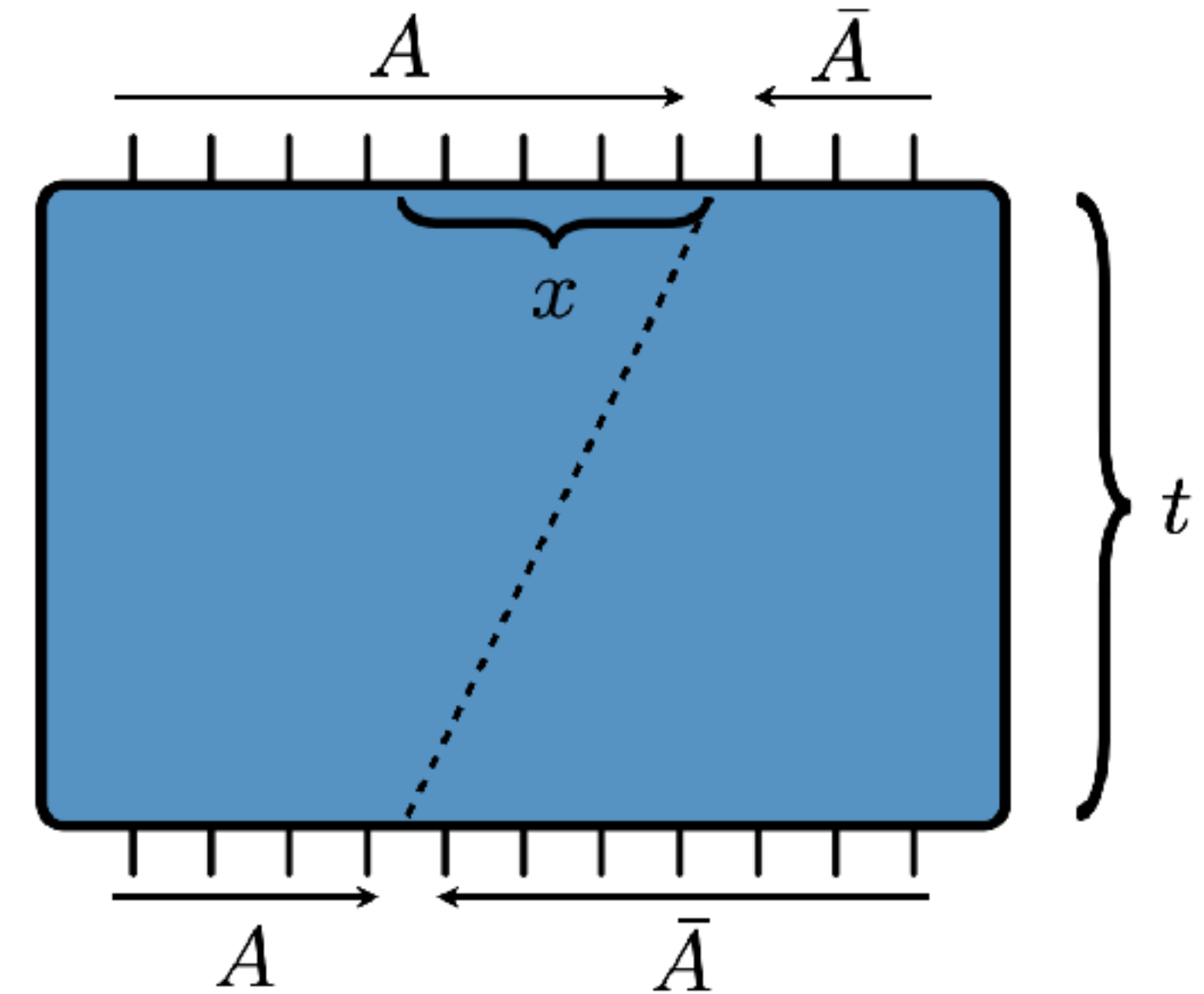
- Microscopic information: entanglement line tension
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- Consistency requires it to be a convex function



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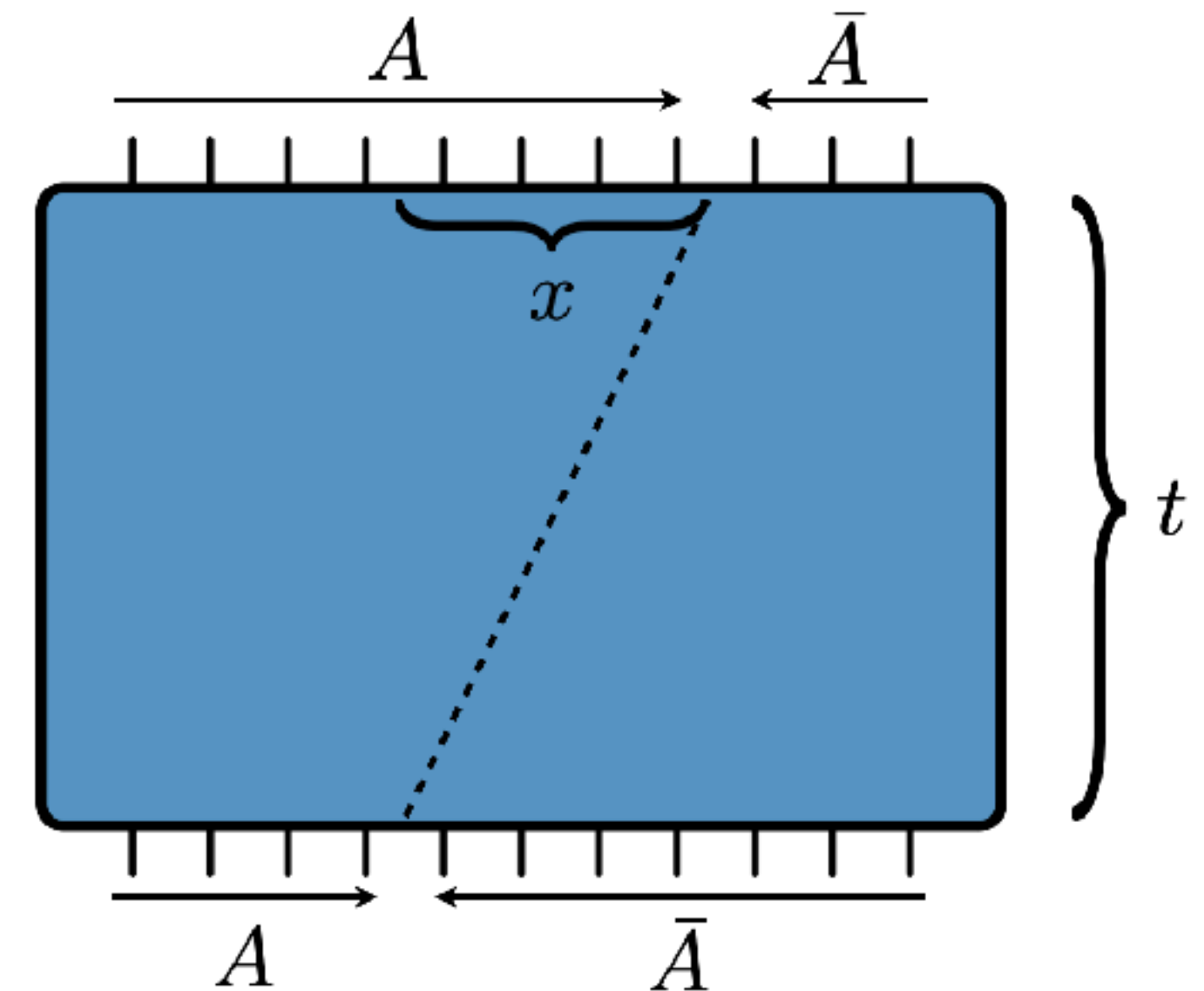
- Force membrane across spacetime



Entanglement Line Tension

- Force membrane across spacetime
- Operator entanglement

$$S_\alpha(x, t) \equiv \frac{1}{1 - \alpha} \log \text{tr} [(\text{tr}_A |\mathcal{U}(t)\rangle \langle \mathcal{U}(t)|)^\alpha]$$

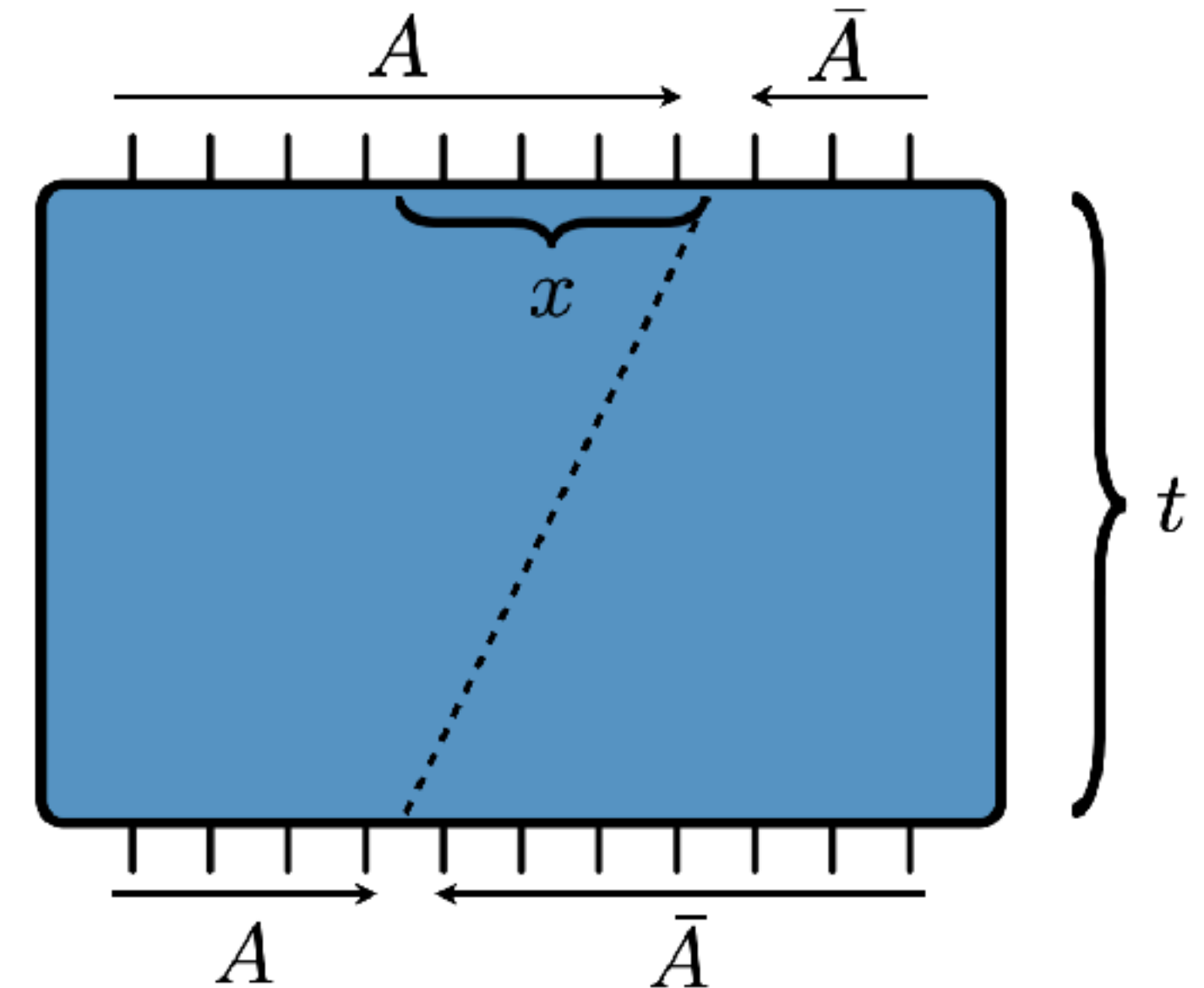


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$$S_\alpha(x, t) \equiv \frac{1}{1 - \alpha} \log \text{tr} [(\text{tr}_A |\mathcal{U}(t)\rangle \langle \mathcal{U}(t)|)^\alpha]$$

- Scales asymptotically as $S_\alpha(x, t) = \log(q) \mathcal{E}_\alpha(v) t$



Entanglement Line Tension

- Introduce

$$(U \otimes U^*)^{\otimes \alpha} = \text{diagram with a central orange square labeled } \alpha \text{ and four lines extending from its corners}$$

$$\text{cup} = q^{-\frac{\alpha}{2}} \overbrace{\text{U} \cdots \text{U}}^{2\alpha},$$

$$\text{cap} = q^{-\frac{\alpha}{2}} \overbrace{\text{U} \cdots \text{U}}^{2\alpha}$$

Entanglement Line Tension

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$$(U \otimes U^*)^{\otimes \alpha} = \text{diagram with a central orange square labeled } \alpha \text{ and four lines extending from it}$$

$$\text{cup with a dot} = q^{-\frac{\alpha}{2}} \overbrace{\text{cup} \cdots \text{cup}}^{2\alpha}, \quad \text{cap with a dot} = q^{-\frac{\alpha}{2}} \overbrace{\text{cap} \cdots \text{cap}}^{2\alpha}$$

- Overlap of permutation states

$$\text{cup with a dot} \text{ connected to } \text{cap with a dot} = q^{-\alpha} \text{ diagram of } \alpha \text{ loops} = \left(\frac{1}{q}\right)^{\alpha-1}$$

Entanglement Line Tension

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$$(U \otimes U^*)^{\otimes \alpha} = \text{Diagram with a central orange square labeled } \alpha \text{ and four external lines}$$

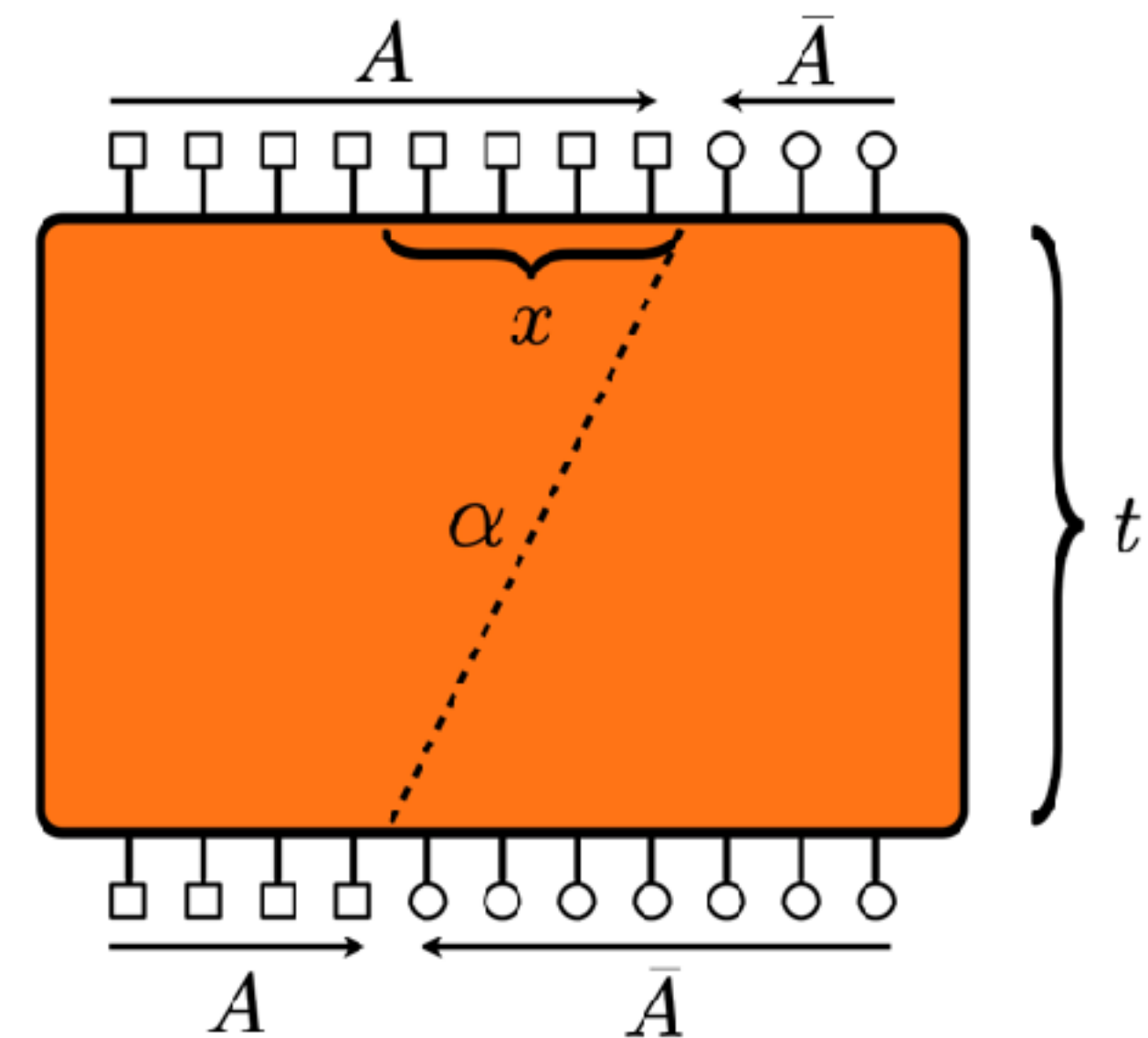
$$\text{Loop with a circle} = q^{-\frac{\alpha}{2}} \overbrace{\text{U} \cdots \text{U}}^{2\alpha}, \quad \text{Loop with a square} = q^{-\frac{\alpha}{2}} \overbrace{\text{U} \cdots \text{U}}^{2\alpha}$$

- Overlap of permutation states

$$\text{Diagram with a circle on top and a square on bottom} = q^{-\alpha} \text{Wavy line diagram} = \left(\frac{1}{q}\right)^{\alpha-1}$$

- Tensor network representation of moment

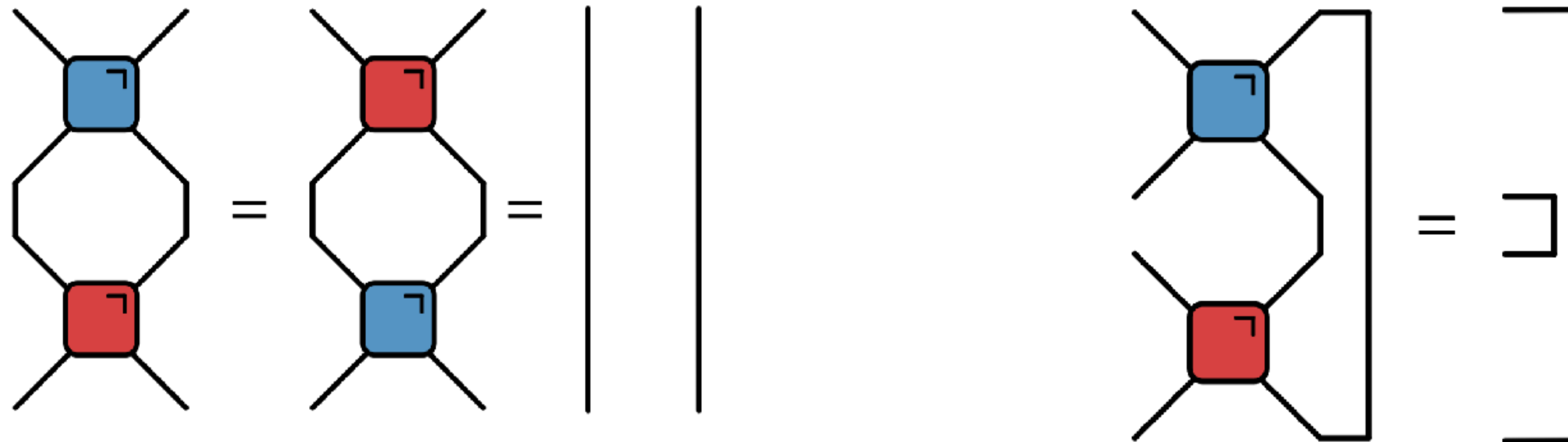
$$Z_{\alpha} = \text{tr} [(\text{tr}_A |\mathcal{U}(t)\rangle \langle \mathcal{U}(t)|)^{\alpha}] =$$



Dual-unitary circuits

- Exactly solvable yet chaotic models: dual-unitary circuits

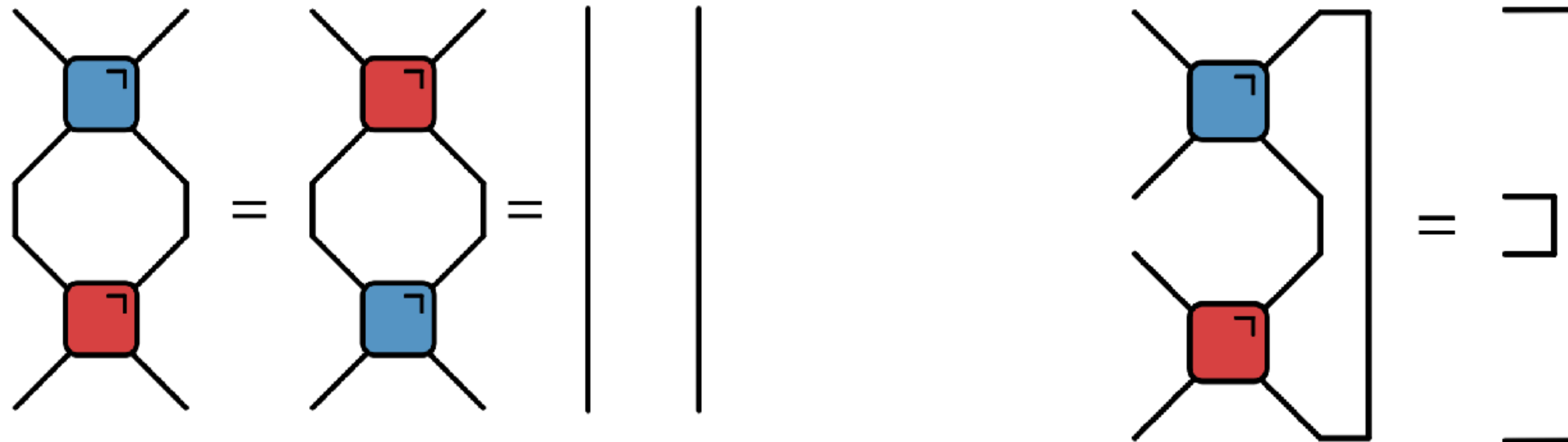
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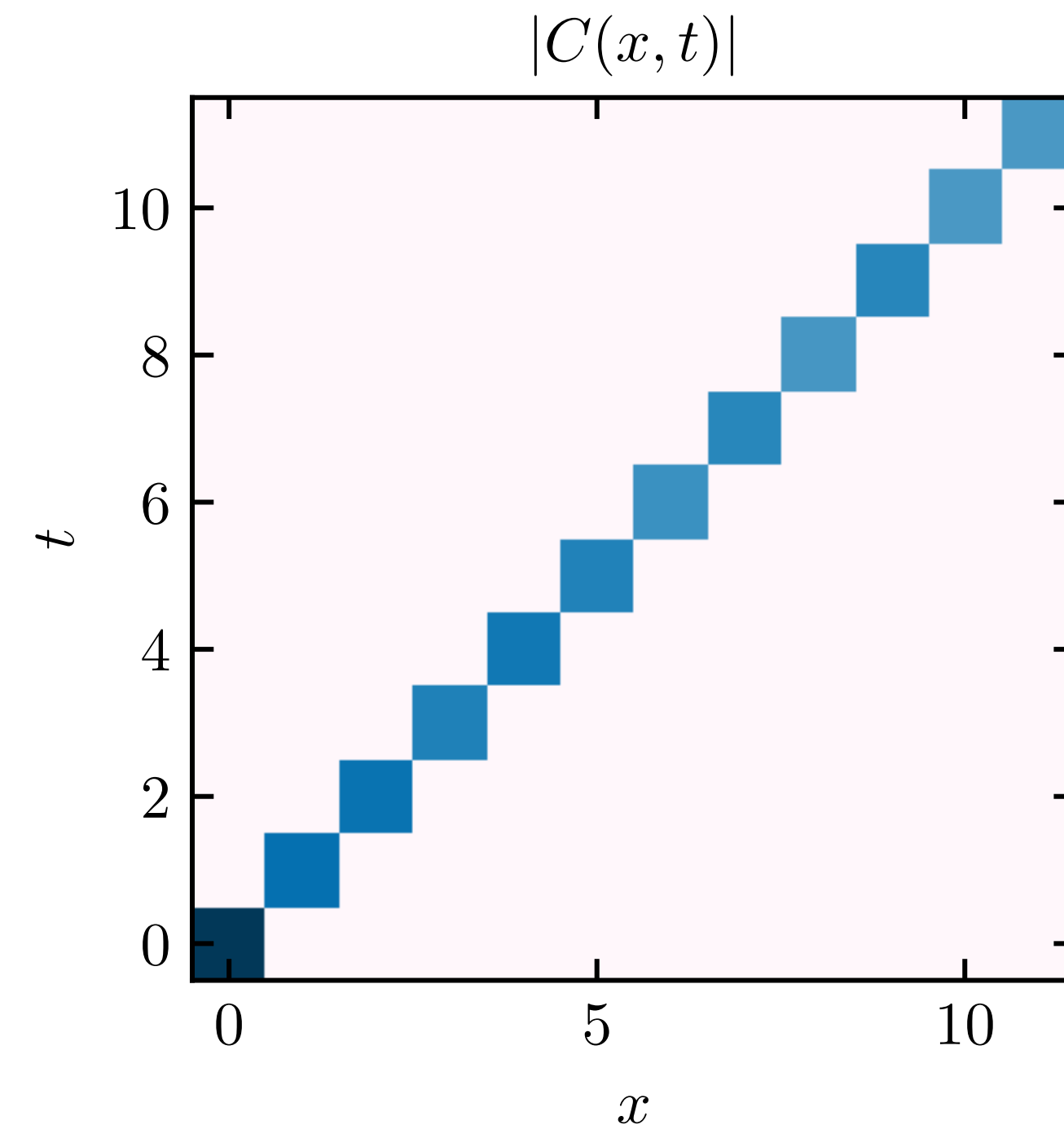


- Simplest example: SWAP gate (non-interacting!)

$$\text{SWAP} = \begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array}$$

Dual-unitary circuits

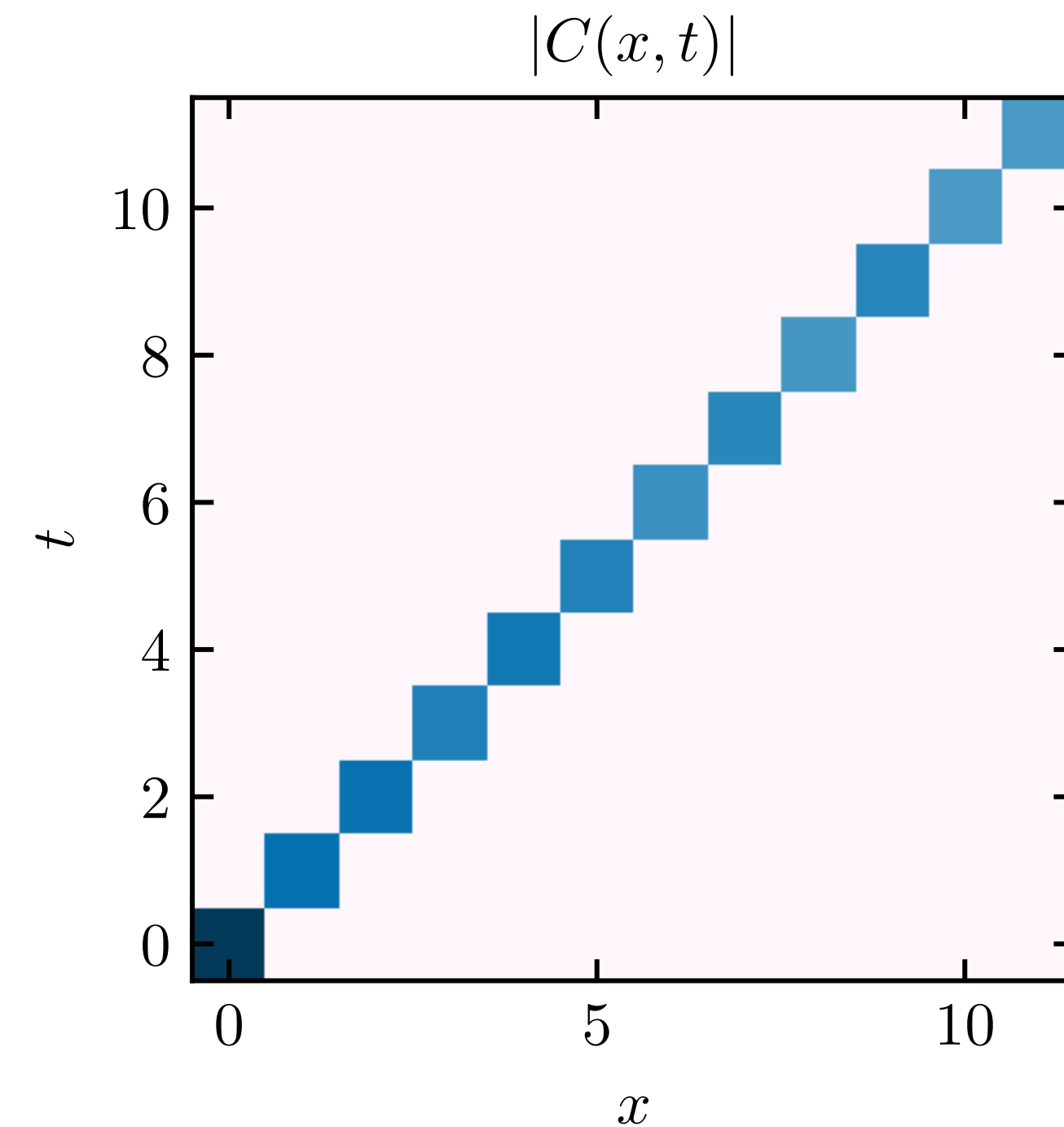
- Correlations on edge of light cone



Bertini, Kos, Prosen (2019),
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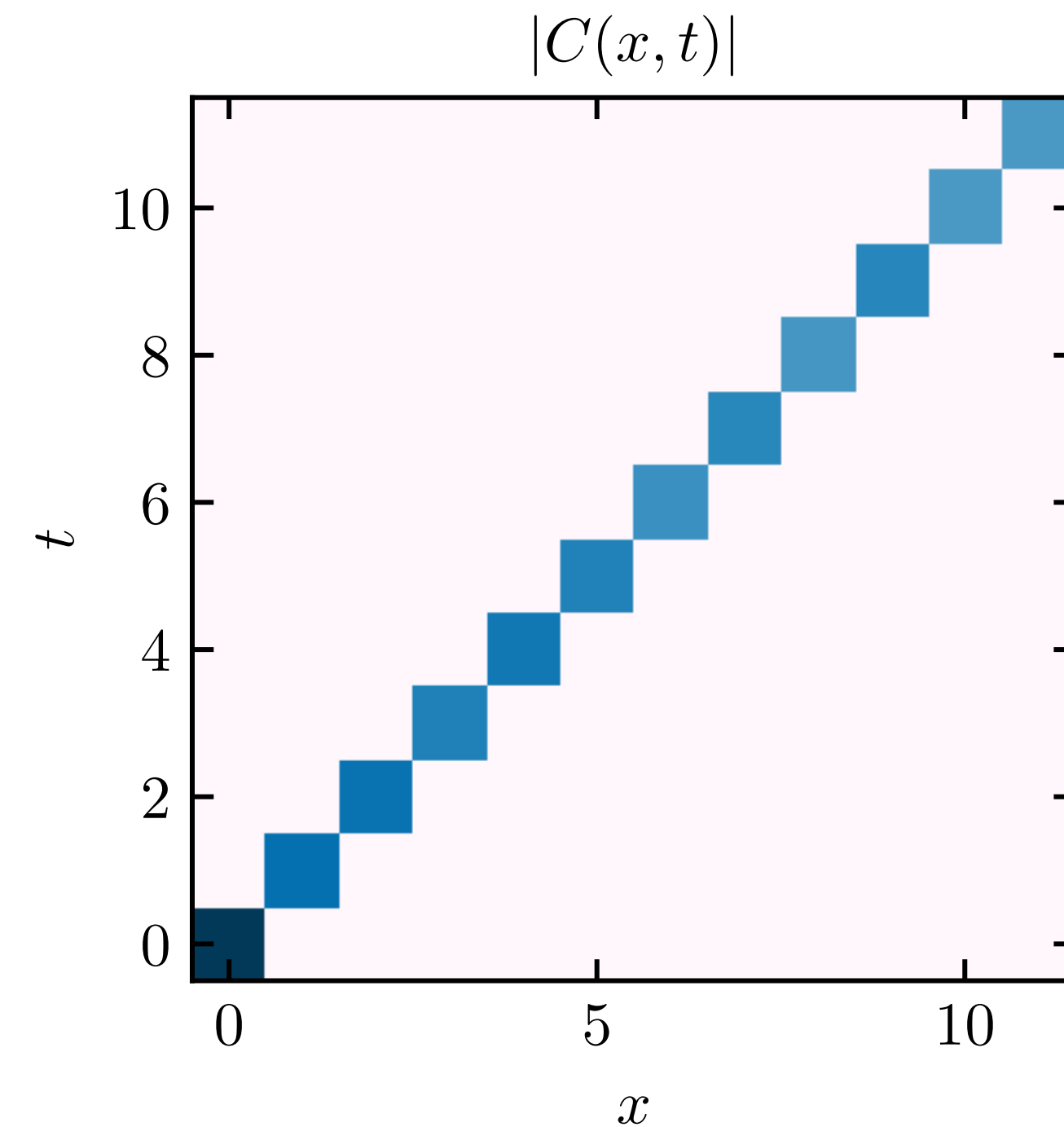
- Correlations on edge of light cone
- Maximal butterfly velocity



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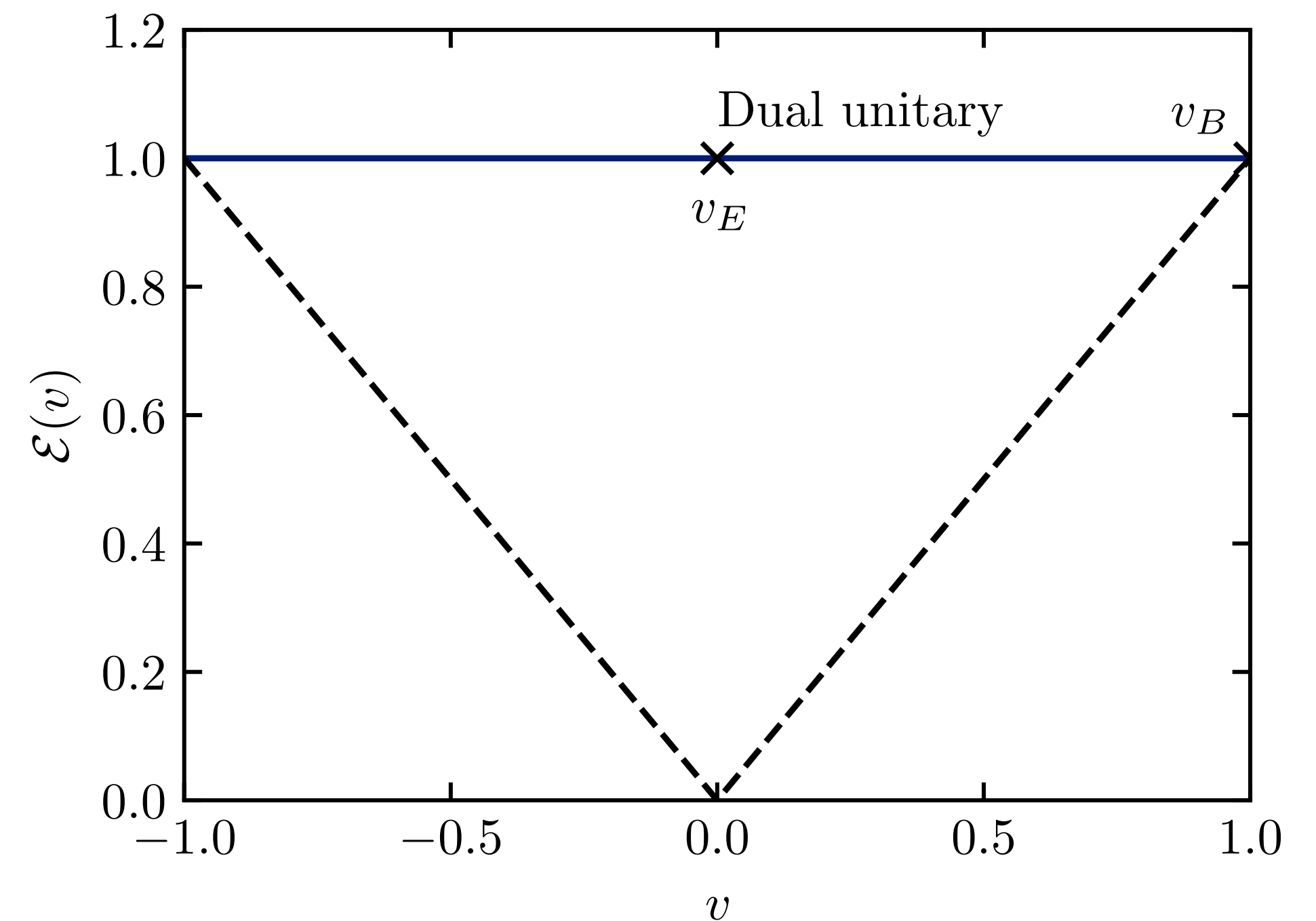
- Correlations on edge of light cone
- Maximal butterfly velocity
- Maximal entanglement growth



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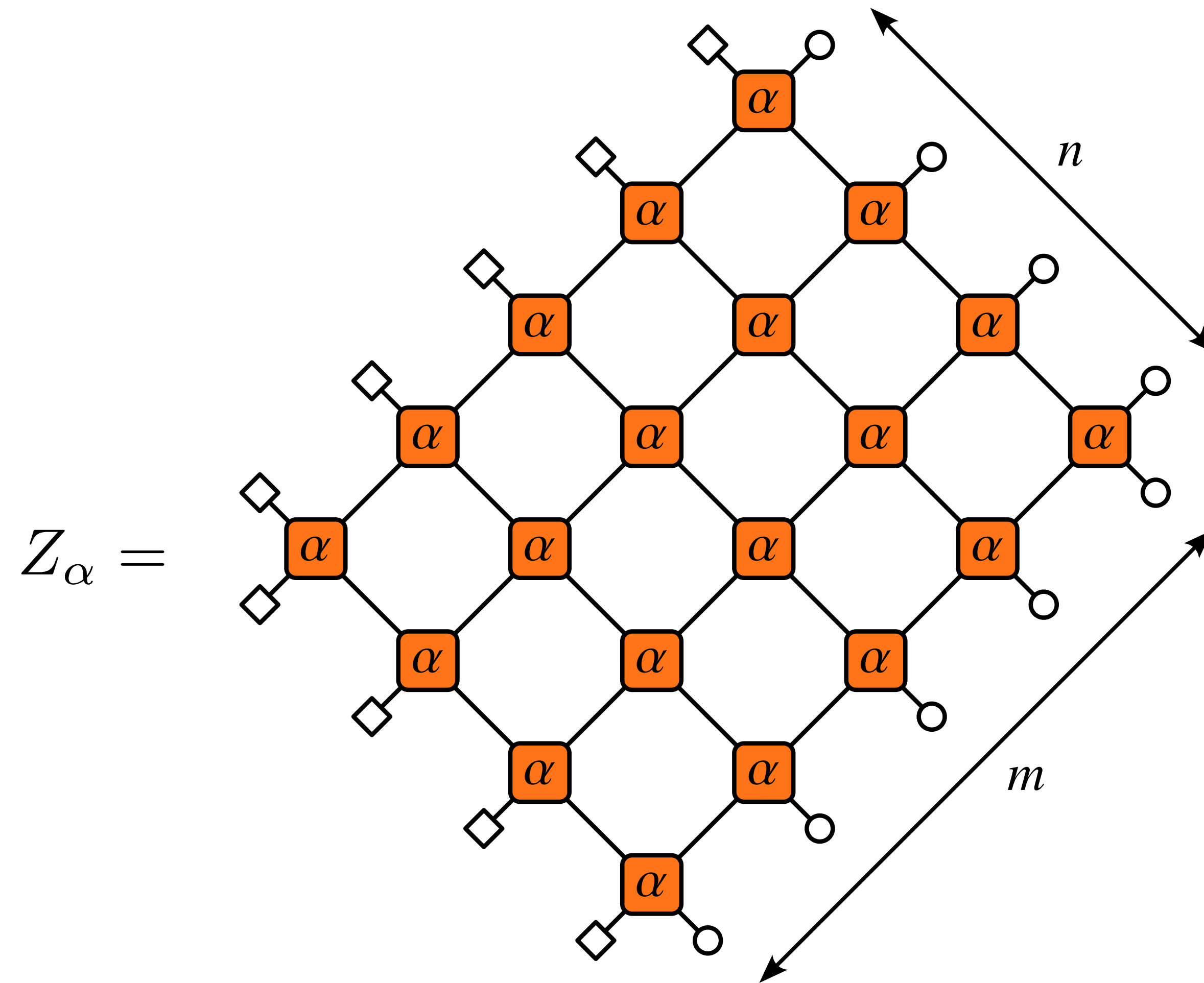
Dual-unitary circuits

- Correlations on edge of light cone
- Maximal butterfly velocity
- Maximal entanglement growth
- Can be derived from flat ELT

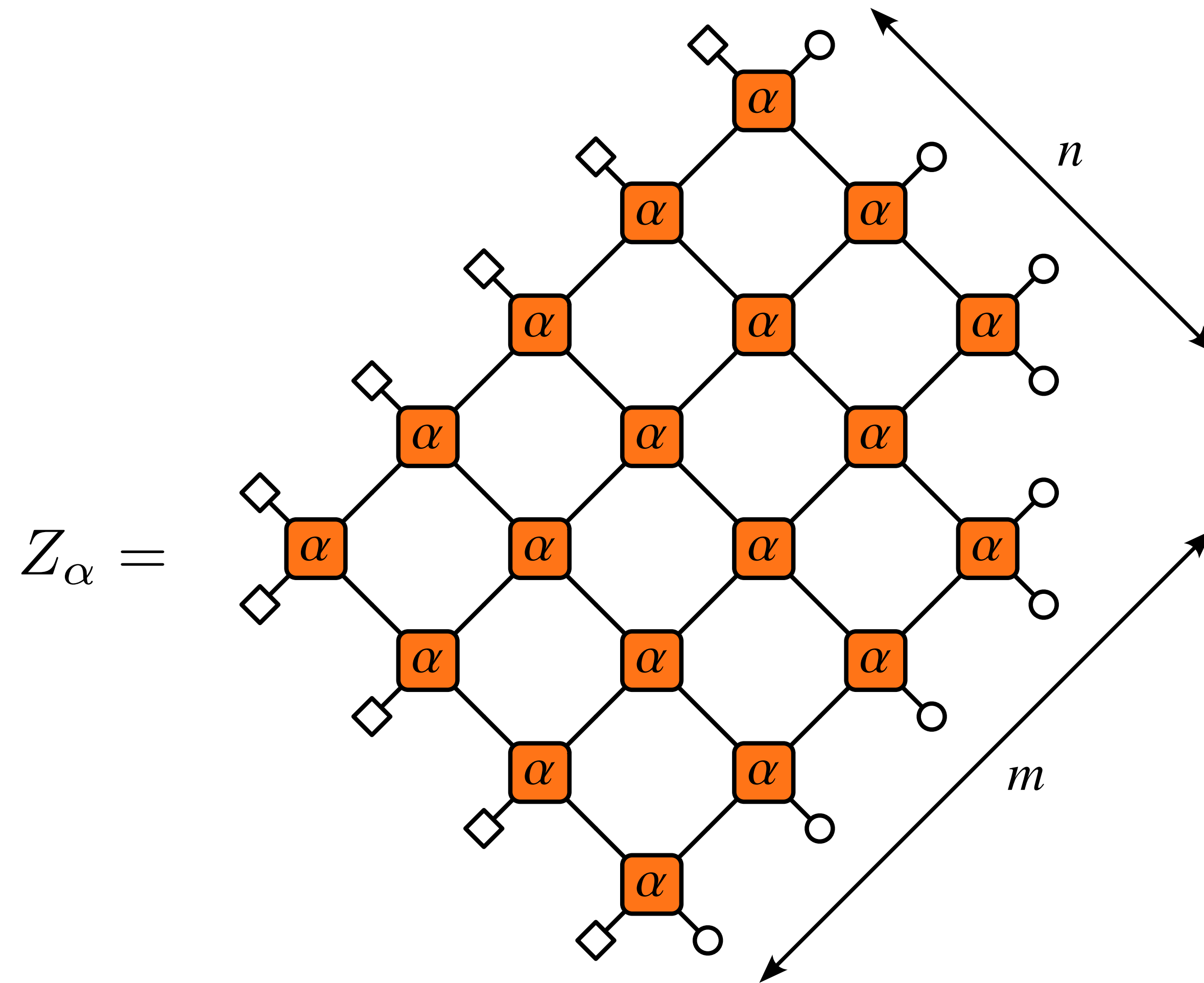


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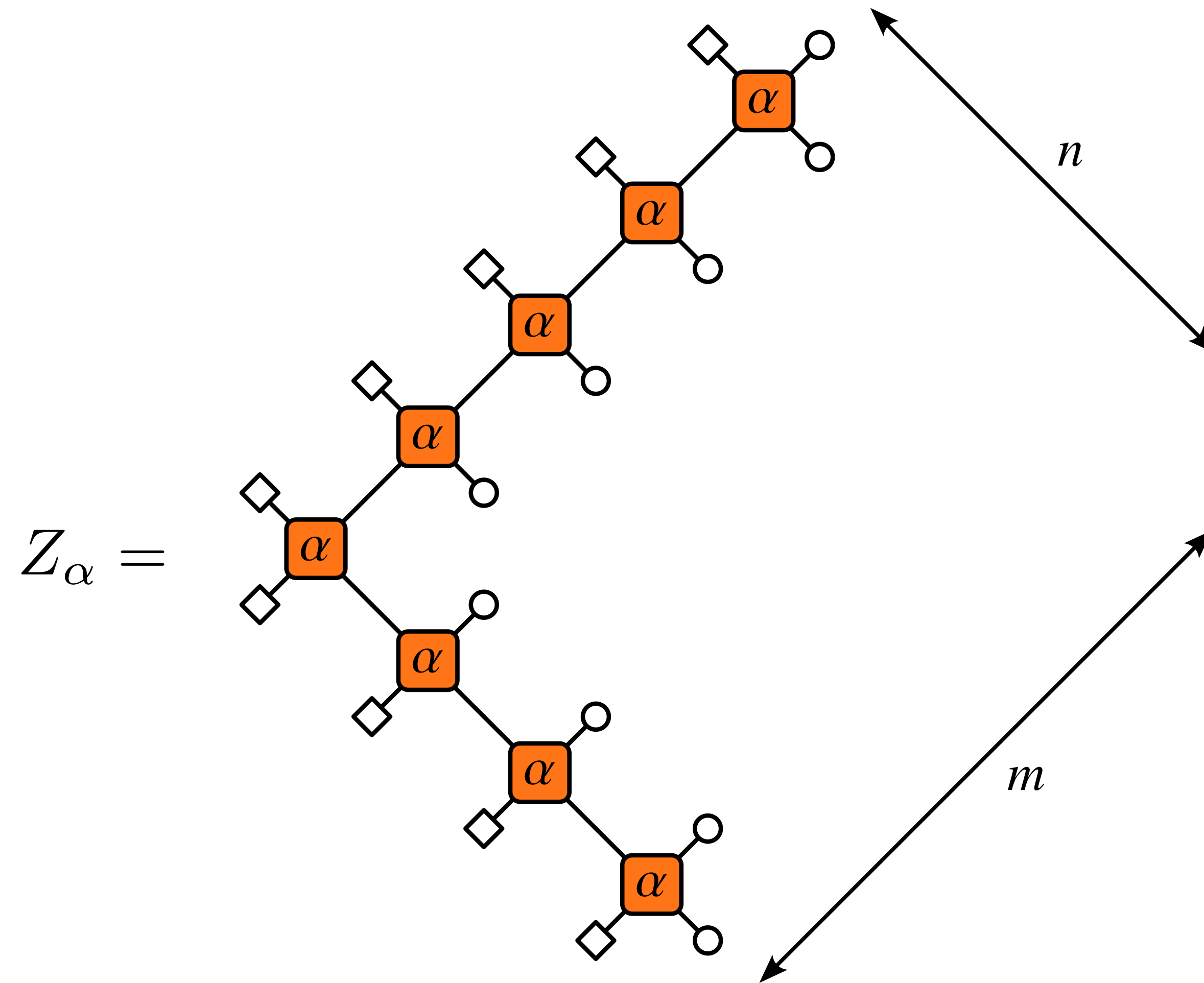
ELT of dual-unitary circuits



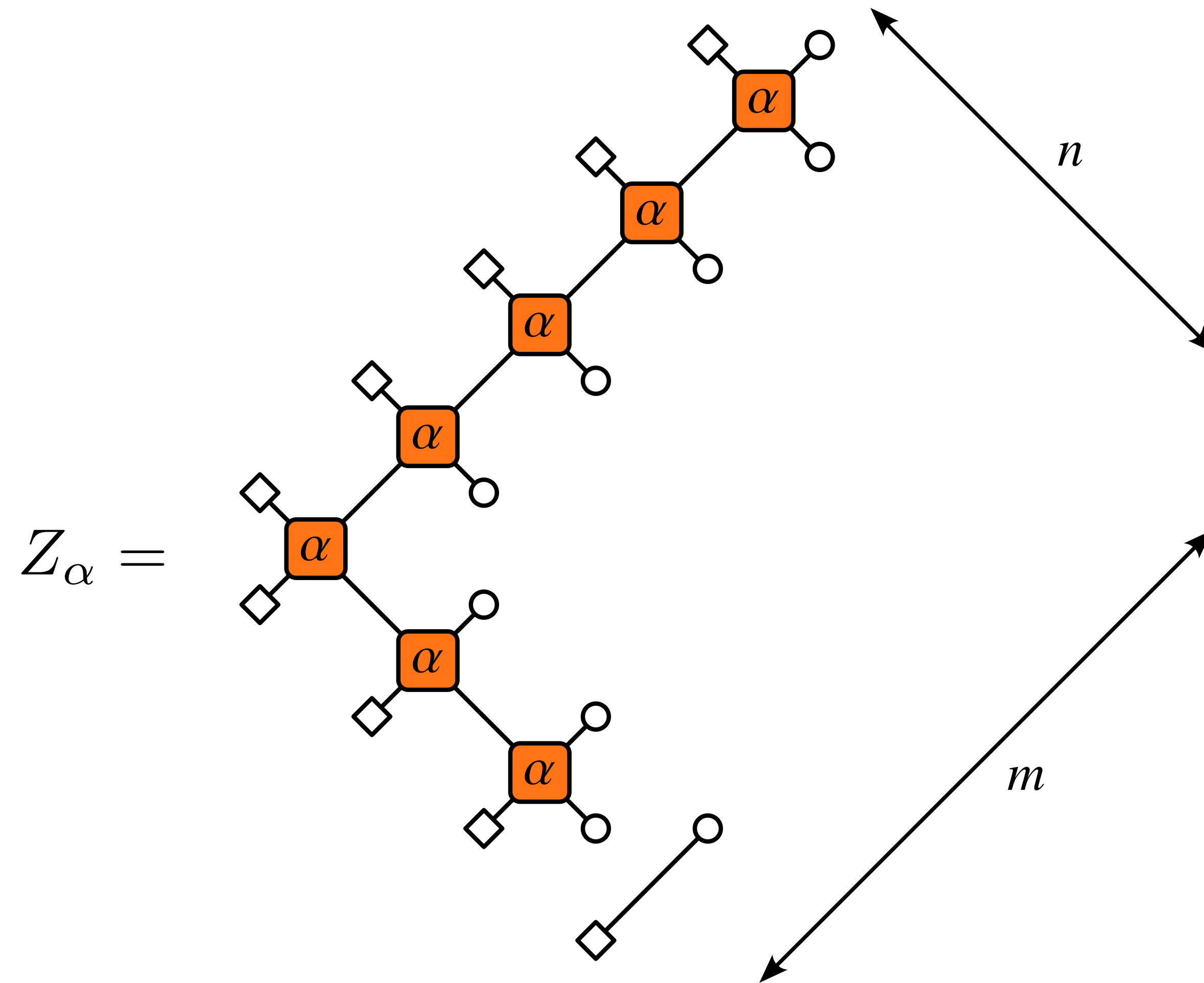
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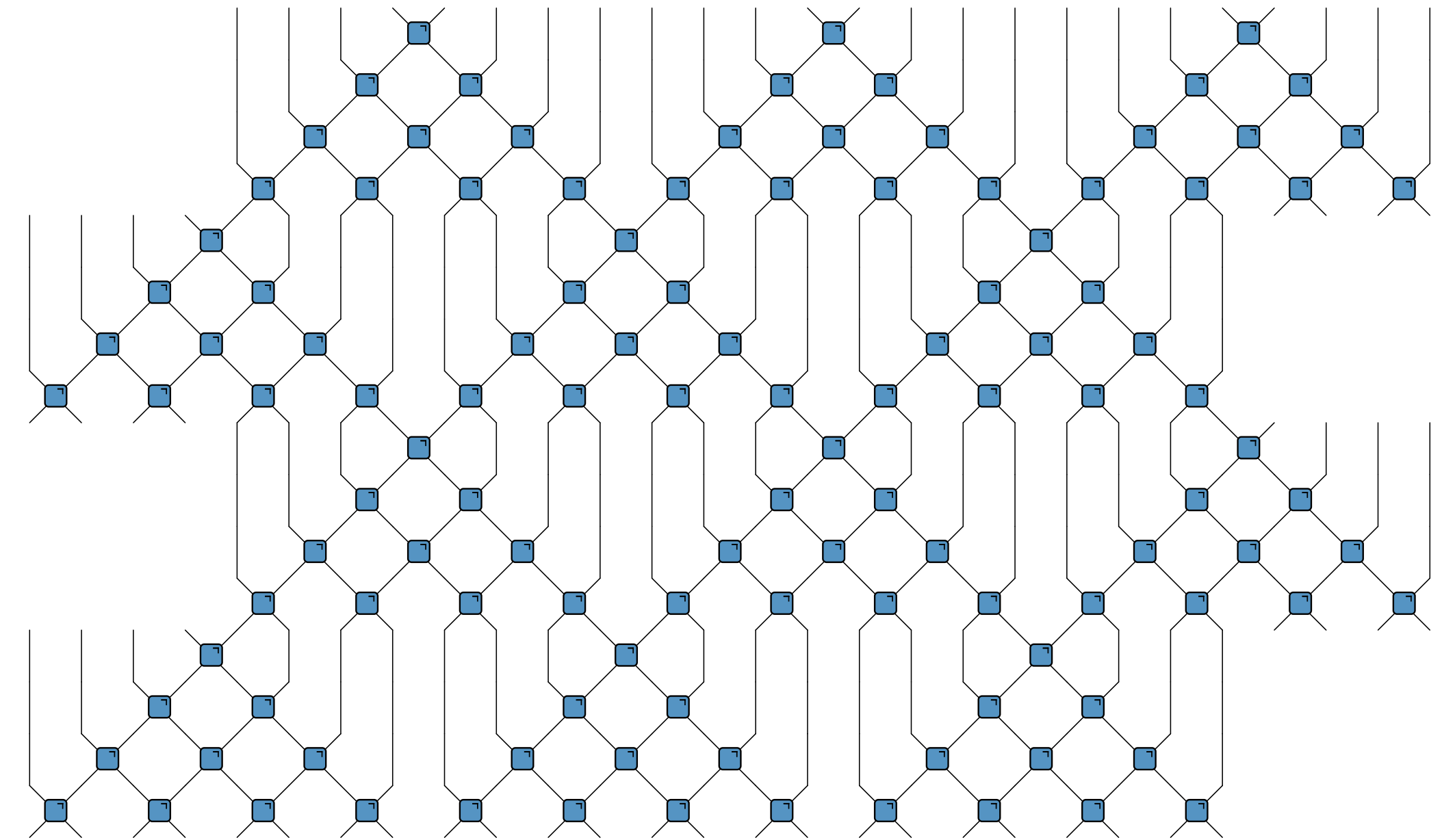


$$Z_{\alpha} = \left(\square \text{---} \circ \right)^{m+n} = q^{(1-\alpha)t}$$

Definition of complete reducibility

A diagram is called completely reducible in a circuit if it can be reduced to a product of overlaps of permutation states.

Examples of completely reducible circuits



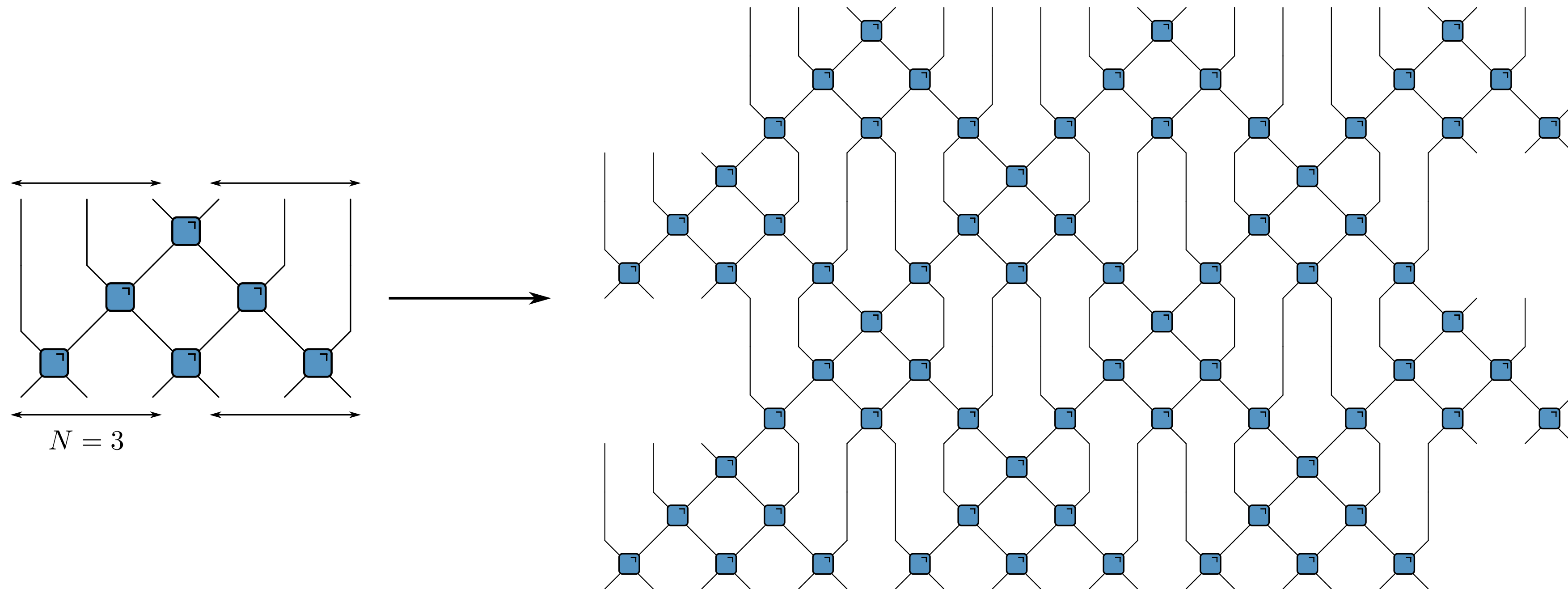
Spacetime lattice circuits



- Keep all interactions dual unitary, but vary circuit geometry

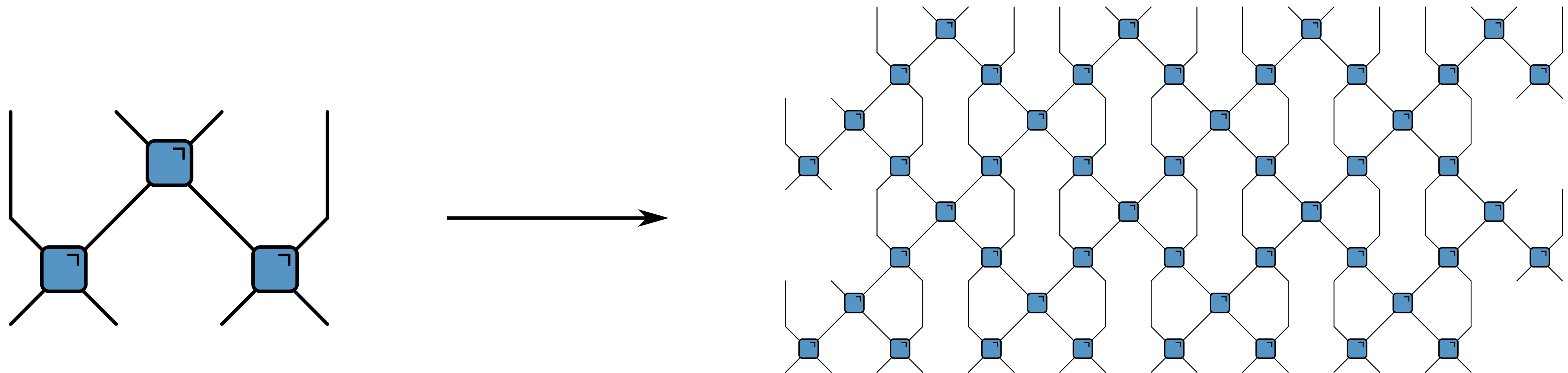
Spacetime lattice circuits

- Keep all interactions dual unitary, but vary circuit geometry
- Build from unit cell — itself a unitary gate



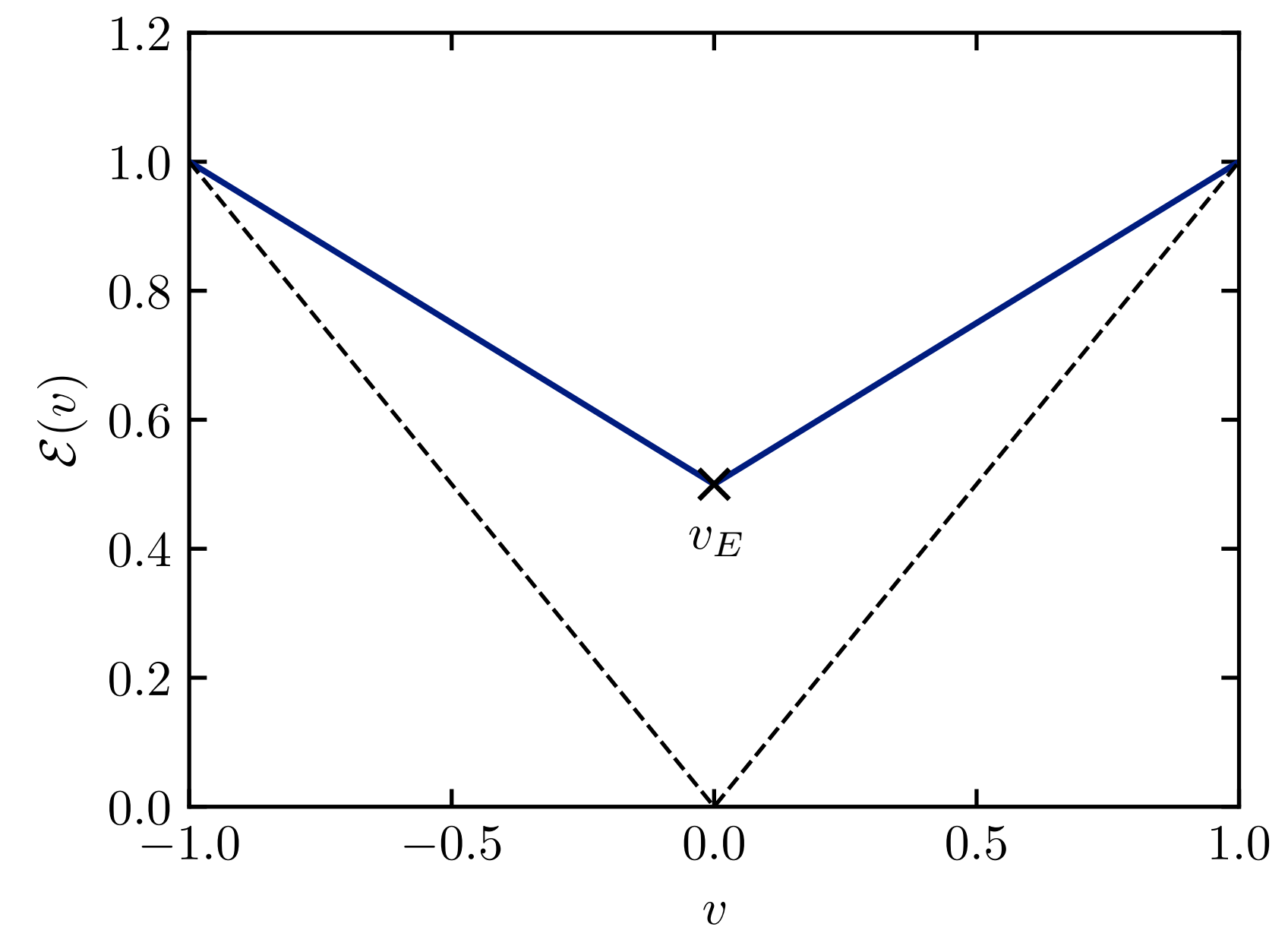
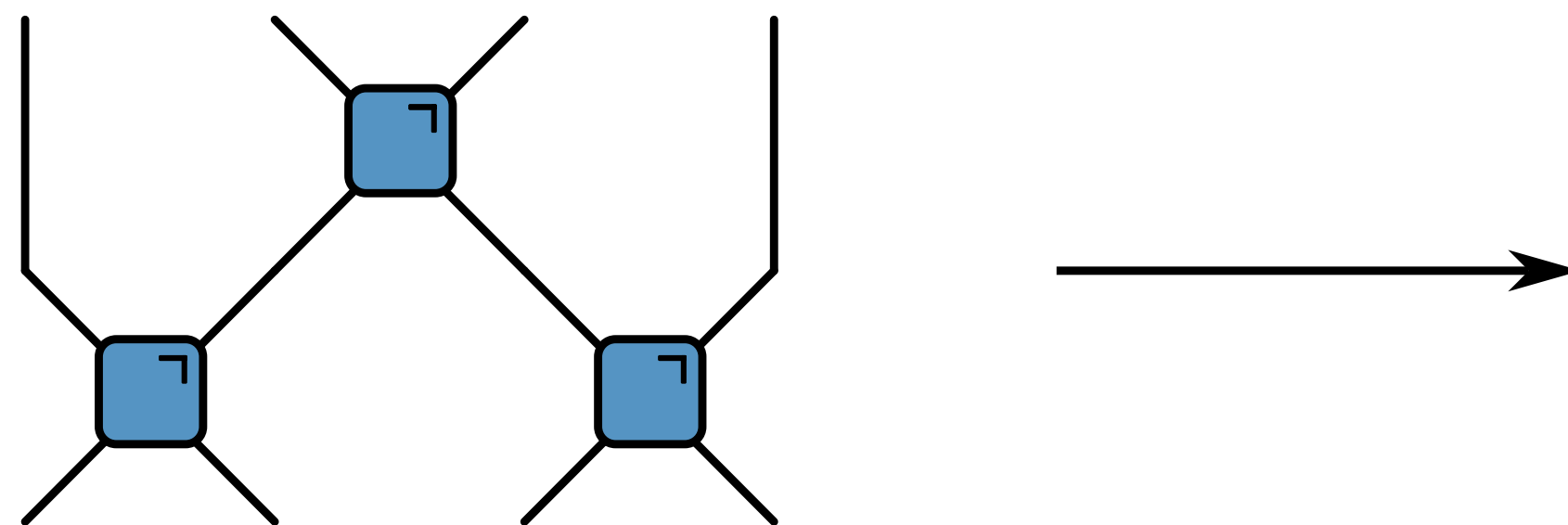
Examples of completely reducible circuits

- Kagome lattice: three rays of information flow



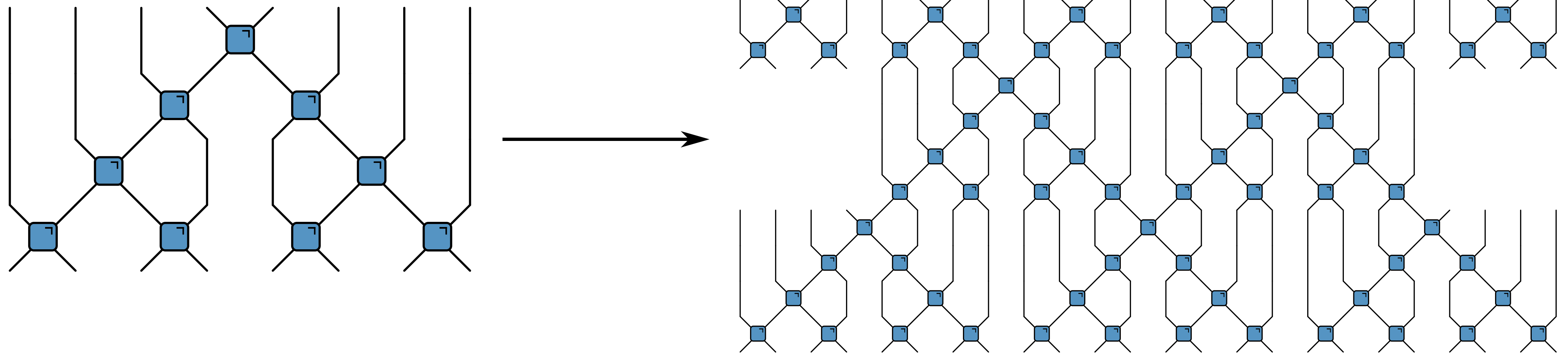
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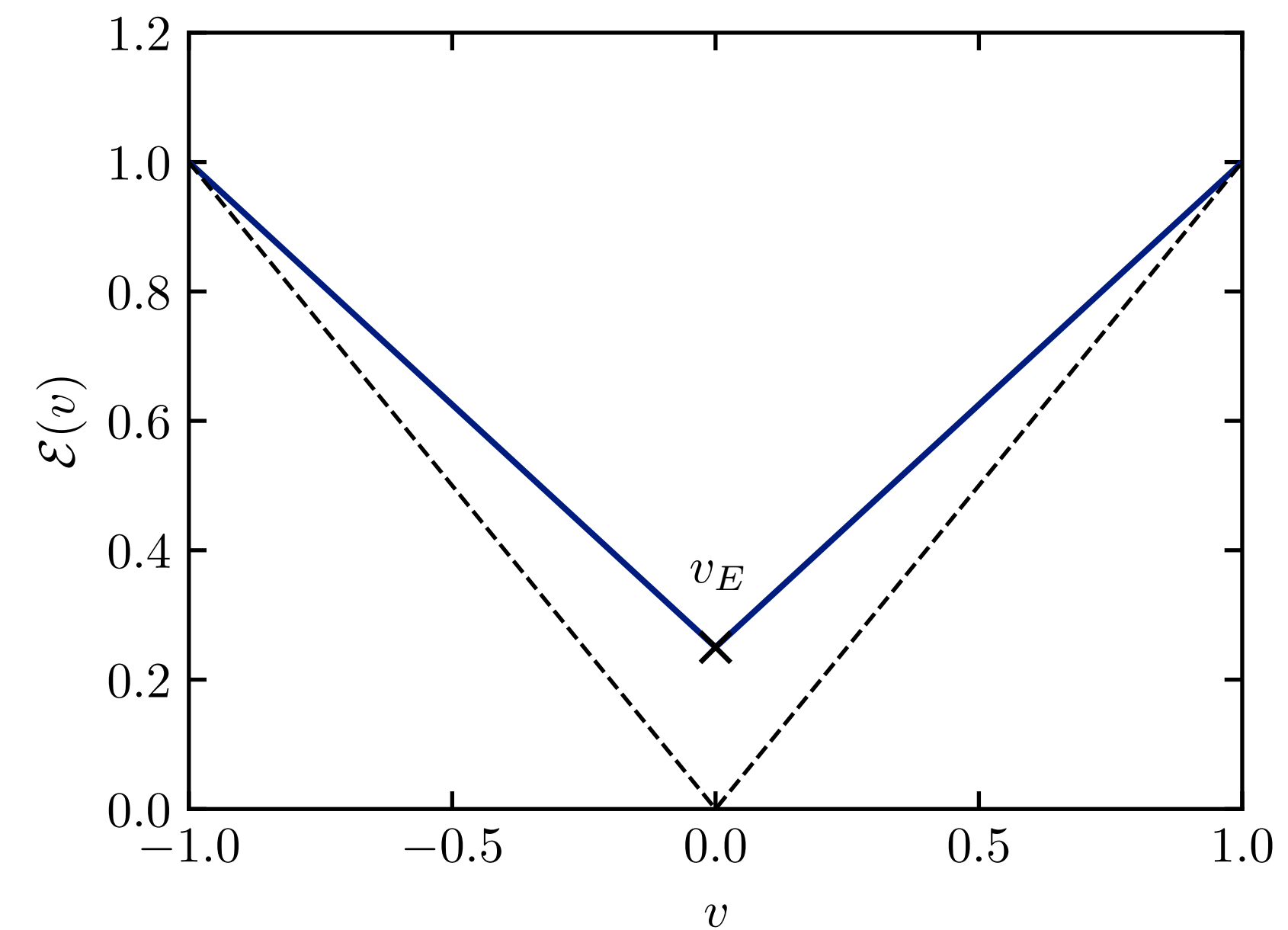
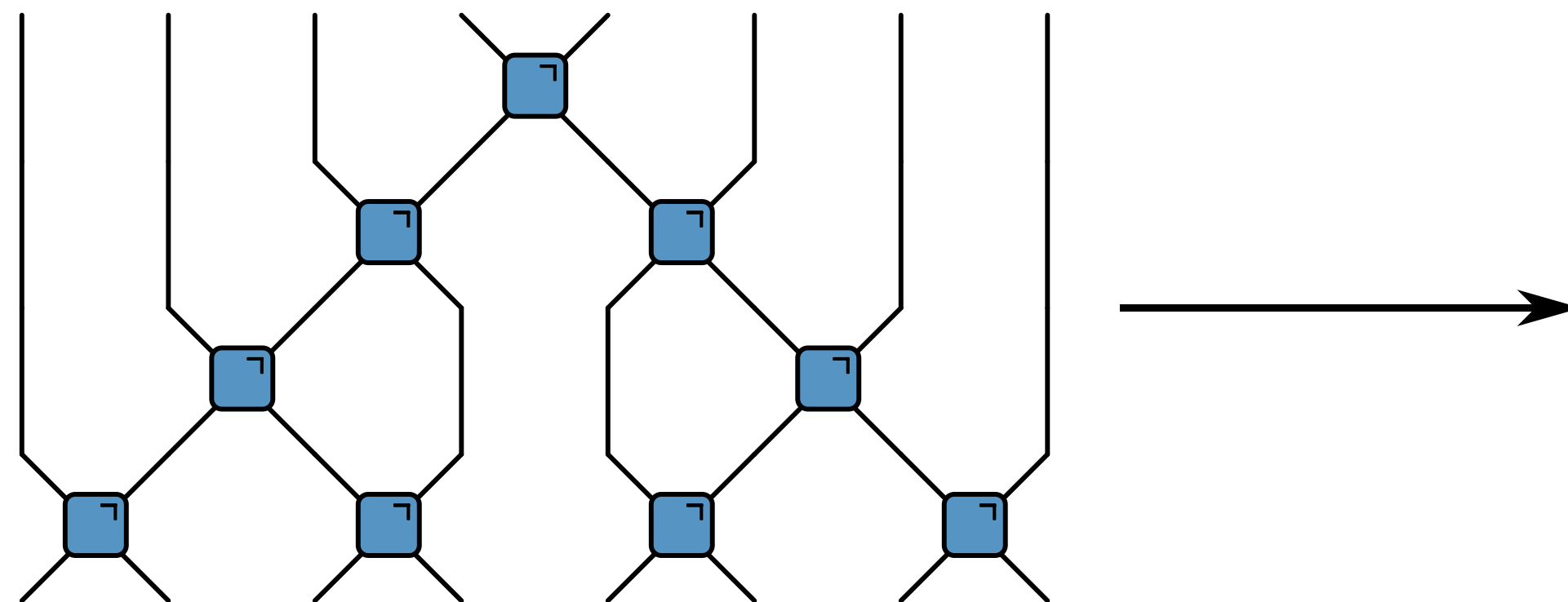
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- Nested Kagome lattice: reduced entanglement velocity
- Breaks sixfold rotational symmetry



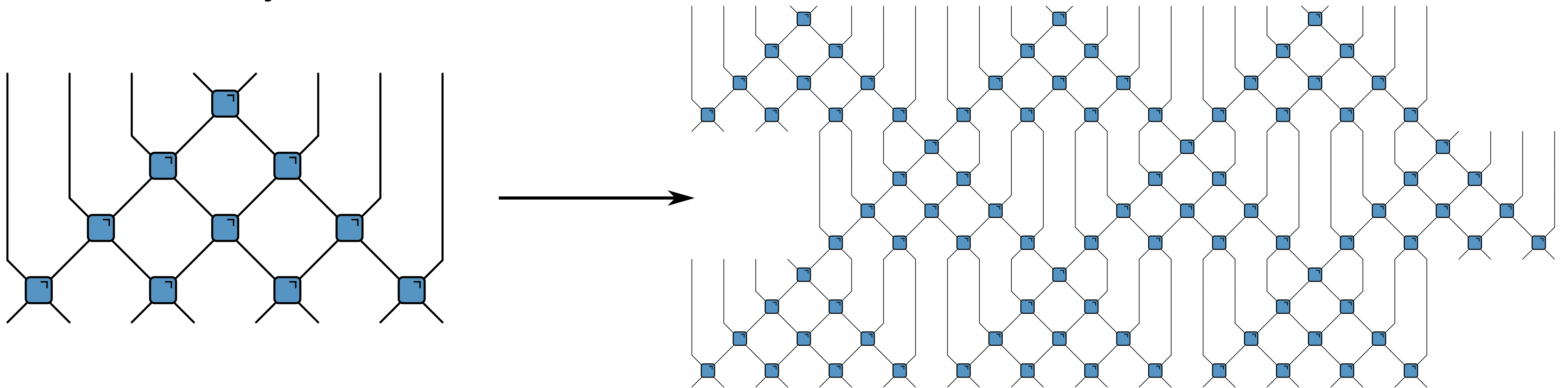
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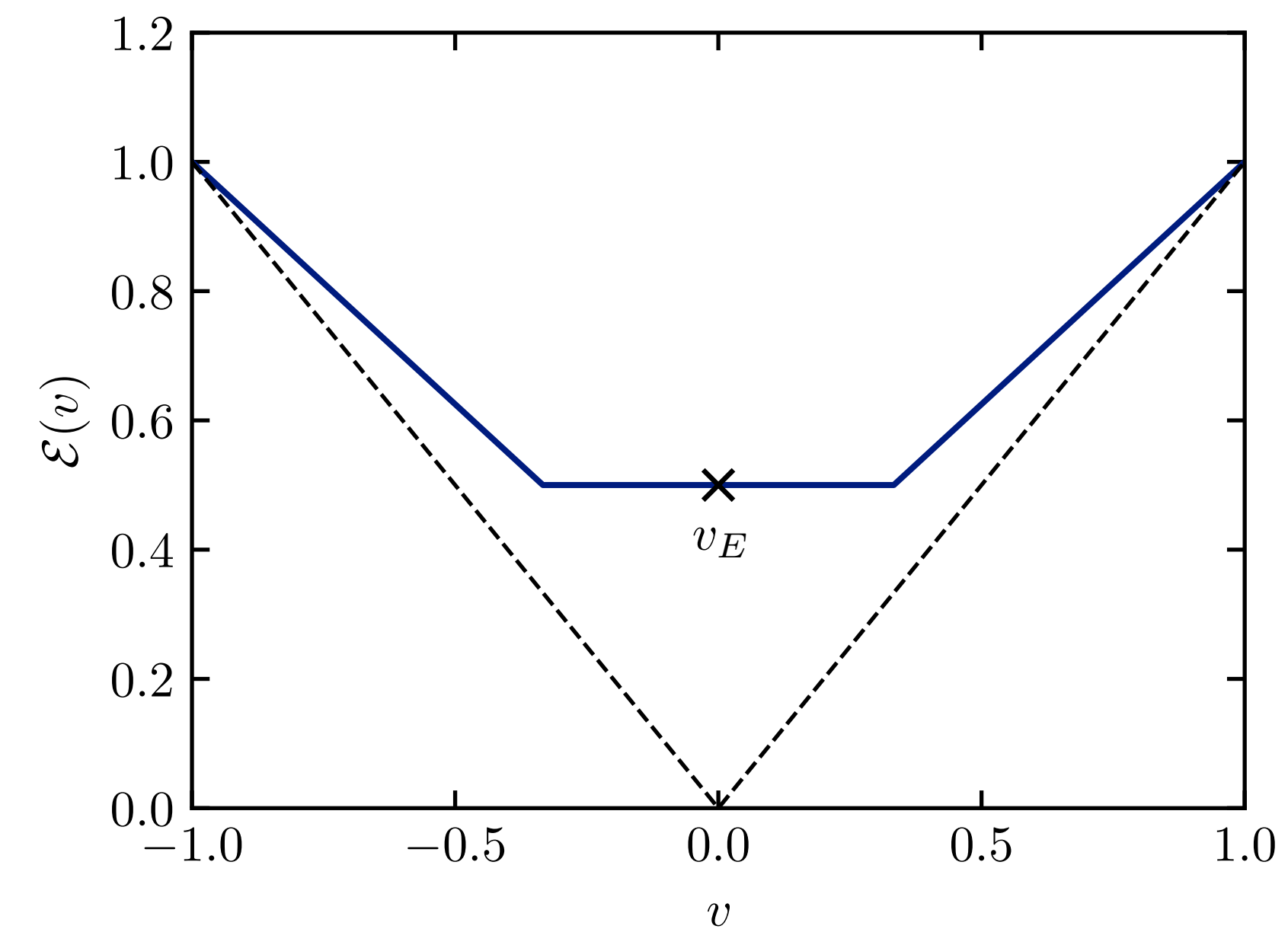
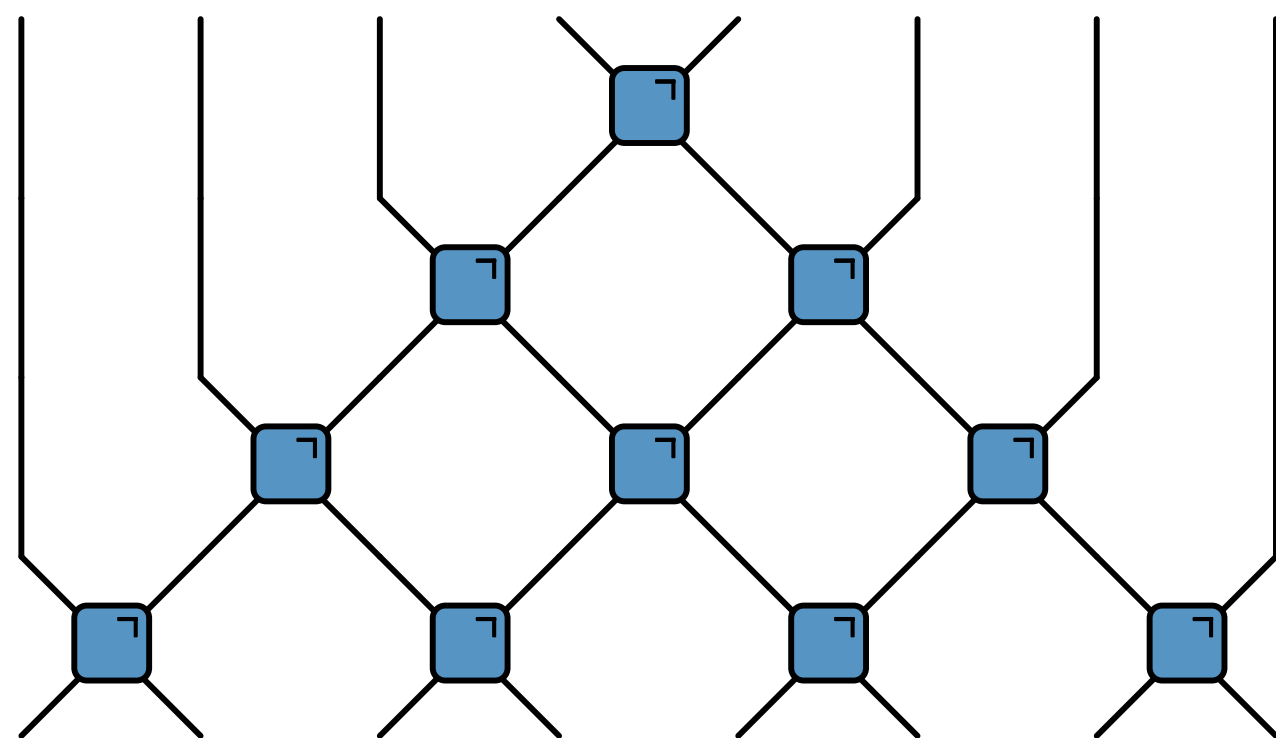
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- Pyramid circuit: four rays of information flow
- No rotational symmetries



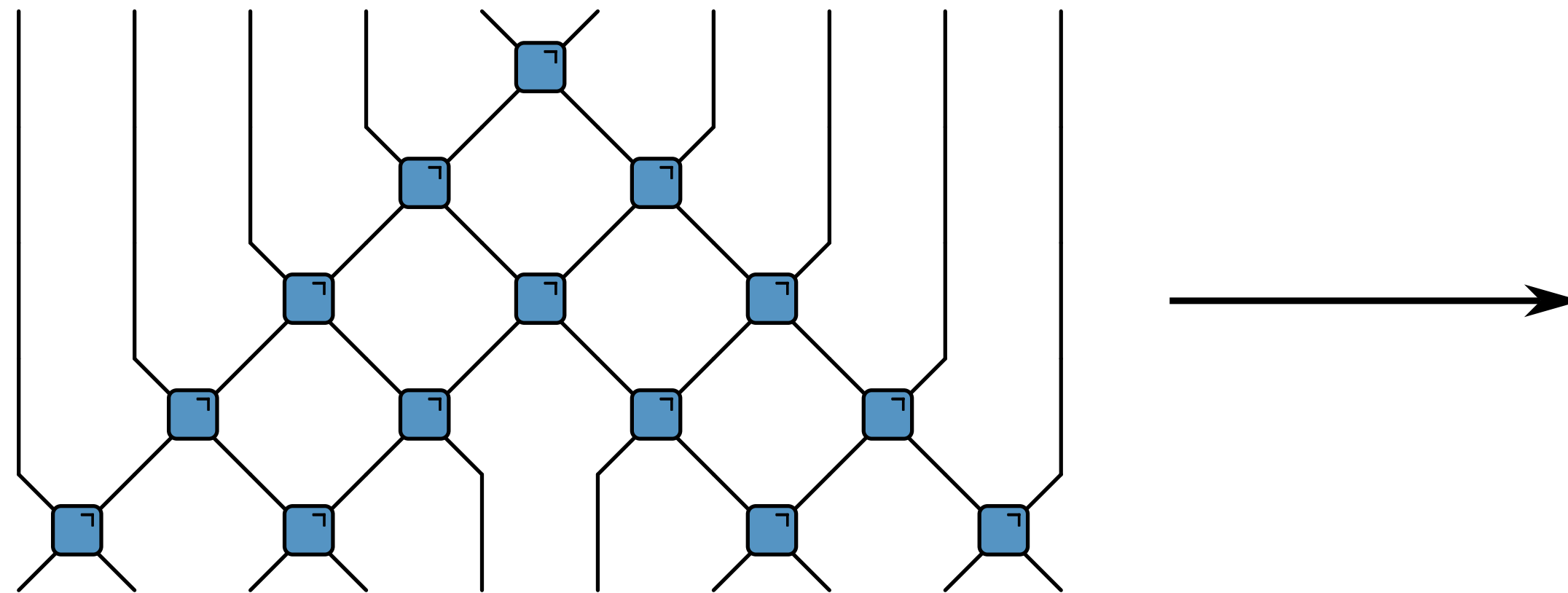
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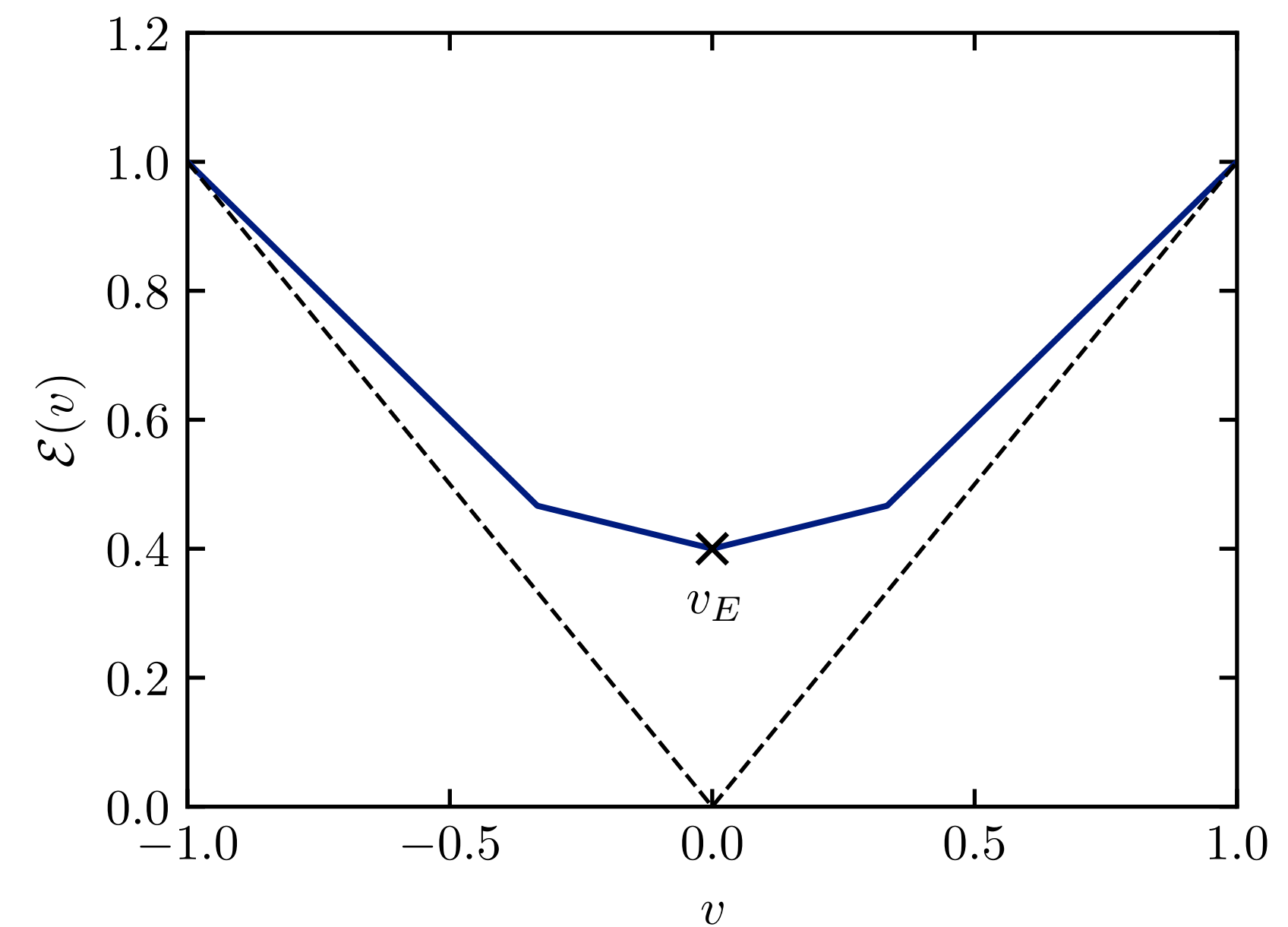
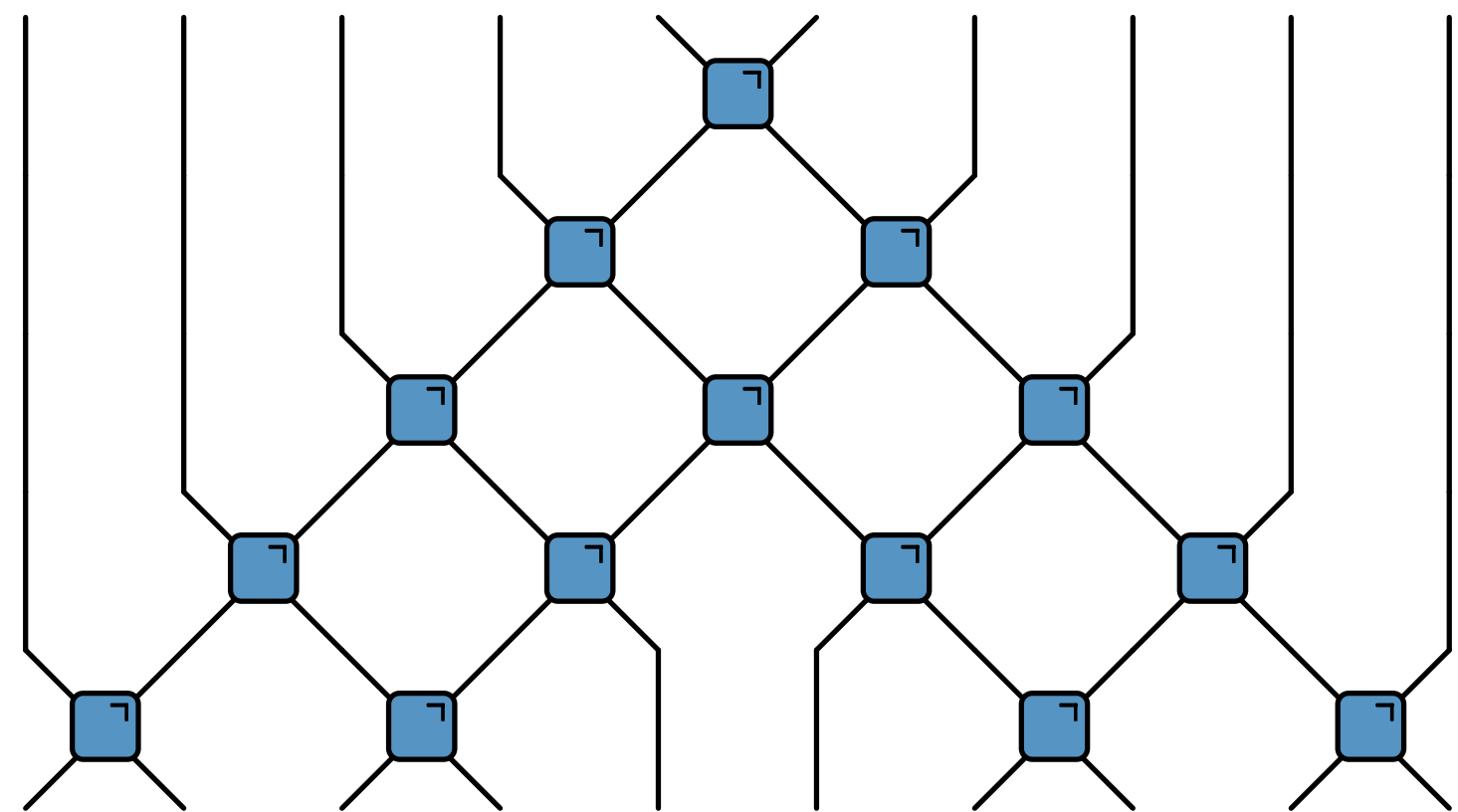
Examples of completely reducible circuits

- Five rays of information flow



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- Five rays of information flow



Consequences of complete reducibility

Flat operator entanglement spectrum

- Diagrams reduce to overlaps of permutations $\rightarrow$ yields power of local Hilbert space dimension

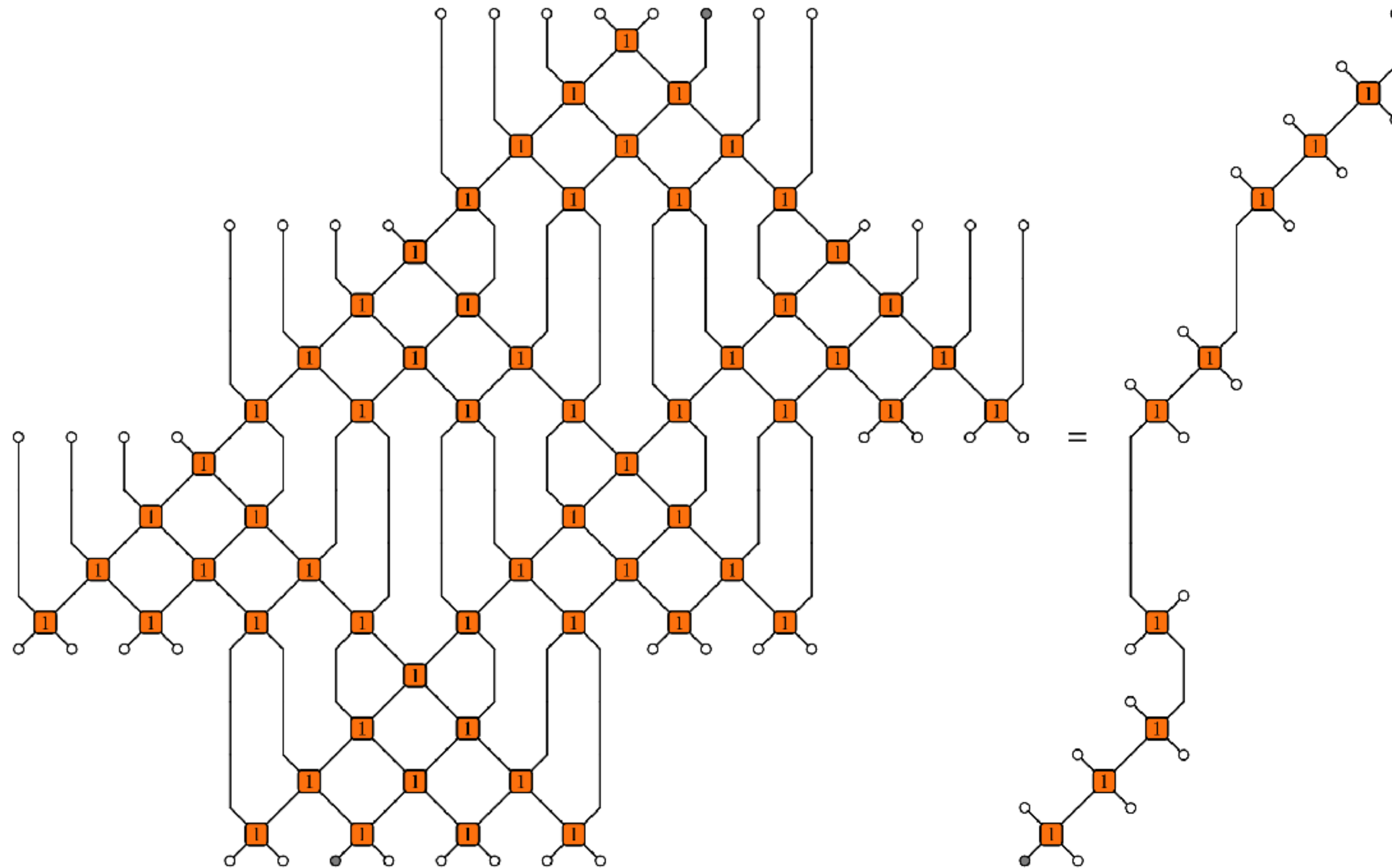
$$Z_{\alpha}(m, n) = \left(\frac{1}{d} \right)^{(\alpha-1)\mathcal{N}}$$

Information flow along discrete directions

- Dynamical correlation functions are only supported along certain discrete worldlines

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- General formula for the ELT: kinks at the velocities of the world lines

$$\mathcal{E}(v) = \sum_{i: v_i < v} \frac{n_i}{2N} (v - v_i) + \sum_{i: v_i > v} \frac{n_i}{2N} (v_i - v), \quad v > 0$$

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- Derive by counting world lines

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- General formula for the ELT: kinks at the velocities of the world lines
- Butterfly velocity: maximal world line velocity
- No diffusive broadening of operator front

Aside: Convexity of the ELT

- What if the ELT formula could be generalized to non-solvable models?

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- Expect continuous density of information flow $n(v)$

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- Relates curvature to information flow density $\rightarrow$ physical reason for convexity

$$\frac{d^2}{dv^2} \mathcal{E}(v) = 2n(v)$$

Entanglement velocity



- Entanglement velocity is quantized

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- Determined by Schmidt rank of unit cell

$$v_E = \frac{\log \mathcal{R}}{\log q^2}$$

Maximal quantum chaos

- Compute spectral form factor numerically and check if conforms to random matrix theory

Maximal quantum chaos

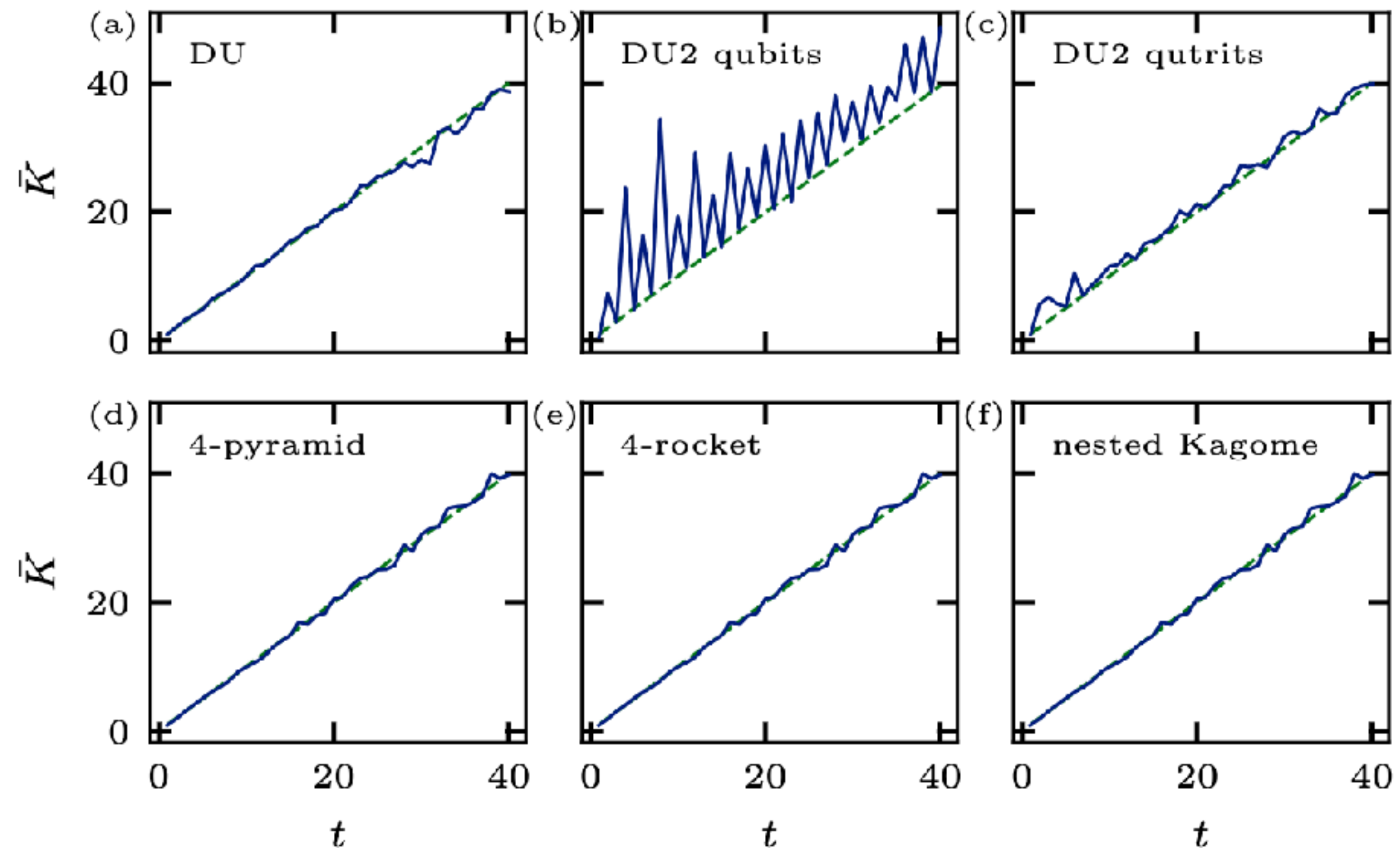
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$$K(t) = \sum_{m,n} e^{i(\varphi_m - \varphi_n)t}$$

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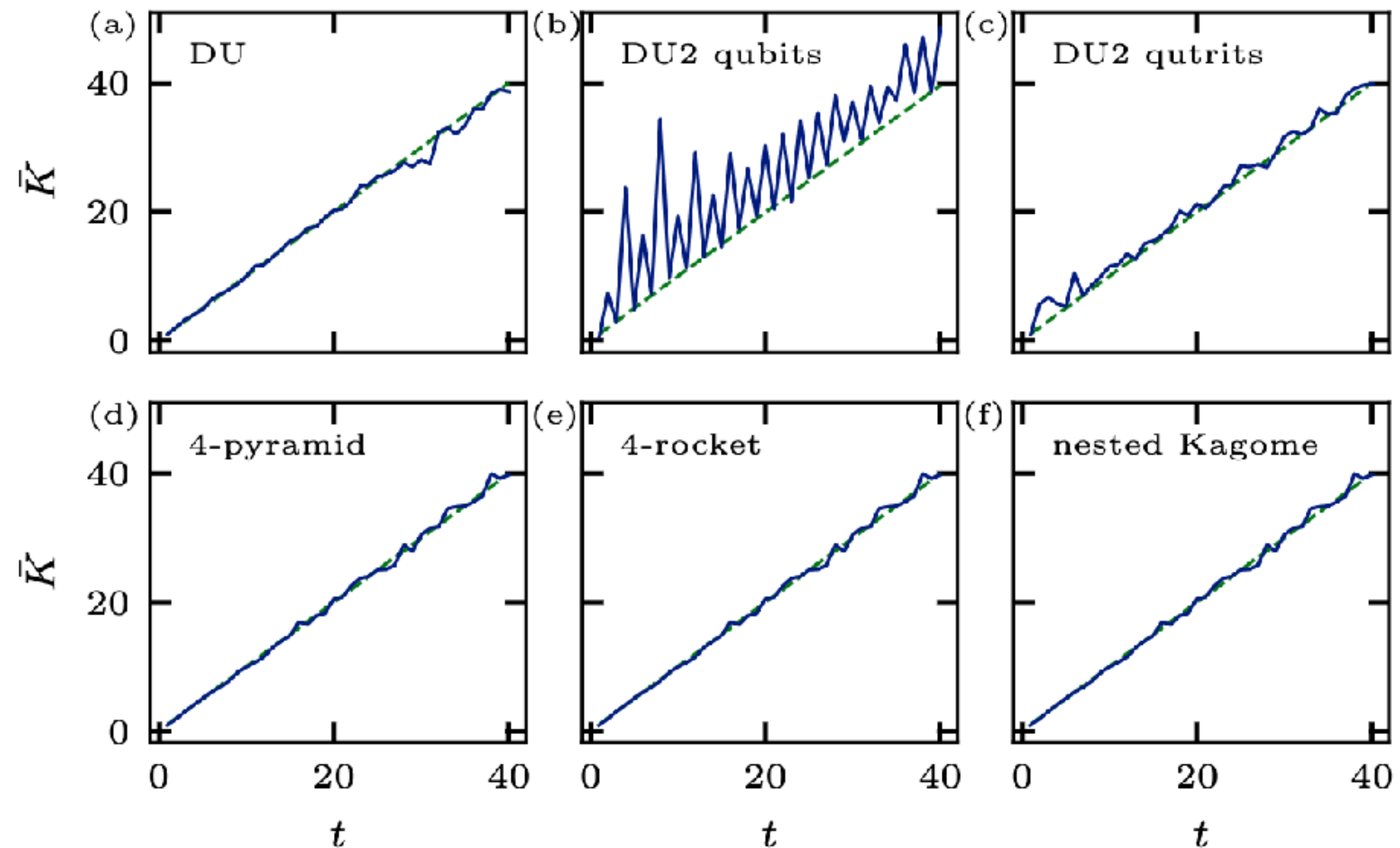
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Maximal quantum chaos

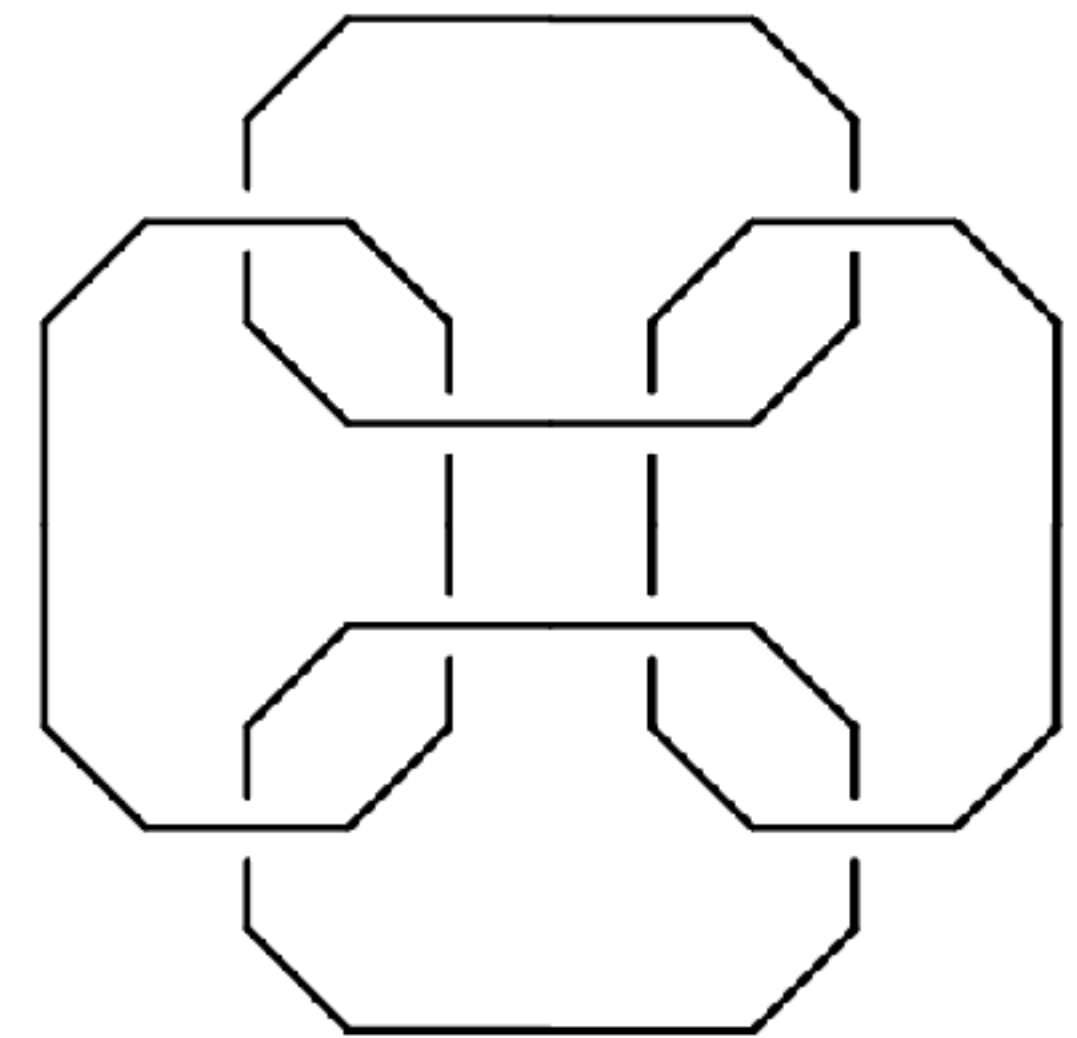
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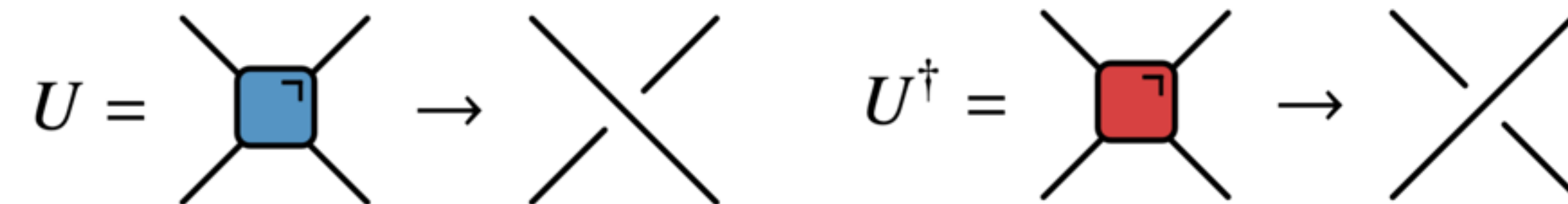
- Absence of characteristic dip $\rightarrow$ Thouless time vanishes

Connection to knot theory



Mapping diagrams to links

- Map gates and conjugate gates to under-/overcrossing of strands



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- Map gates and conjugate gates to under-/overcrossing of strands

$$U = \text{blue gate} \rightarrow \text{crossing} \quad U^\dagger = \text{red gate} \rightarrow \text{crossing}$$

- Unitarity + dual unitarity yields Reidemeister II move

$$\begin{array}{c} \text{blue gate} \\ \text{red gate} \end{array} \rightarrow \text{crossing} = \text{parallel lines} \quad \begin{array}{c} \text{blue gate} \\ \text{red gate} \end{array} \rightarrow \text{crossing} = \text{cup}$$

Complete reducibility $\rightarrow$ trivial links



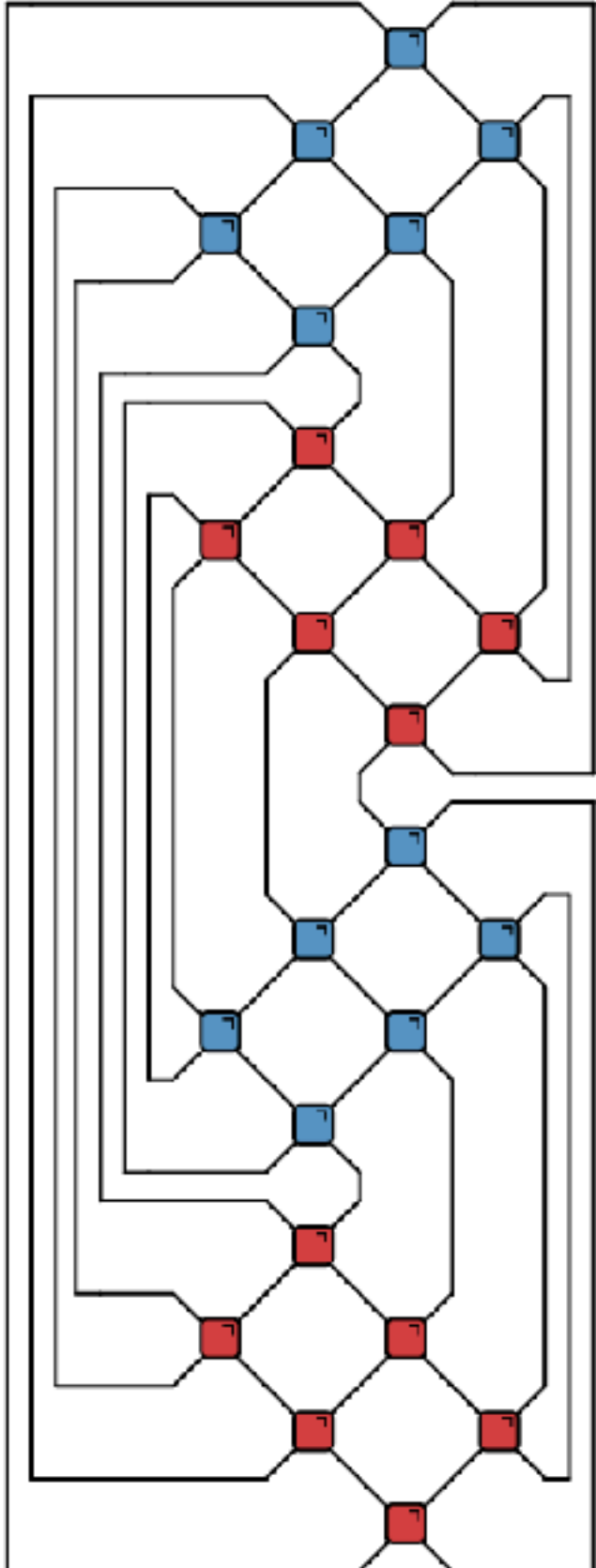
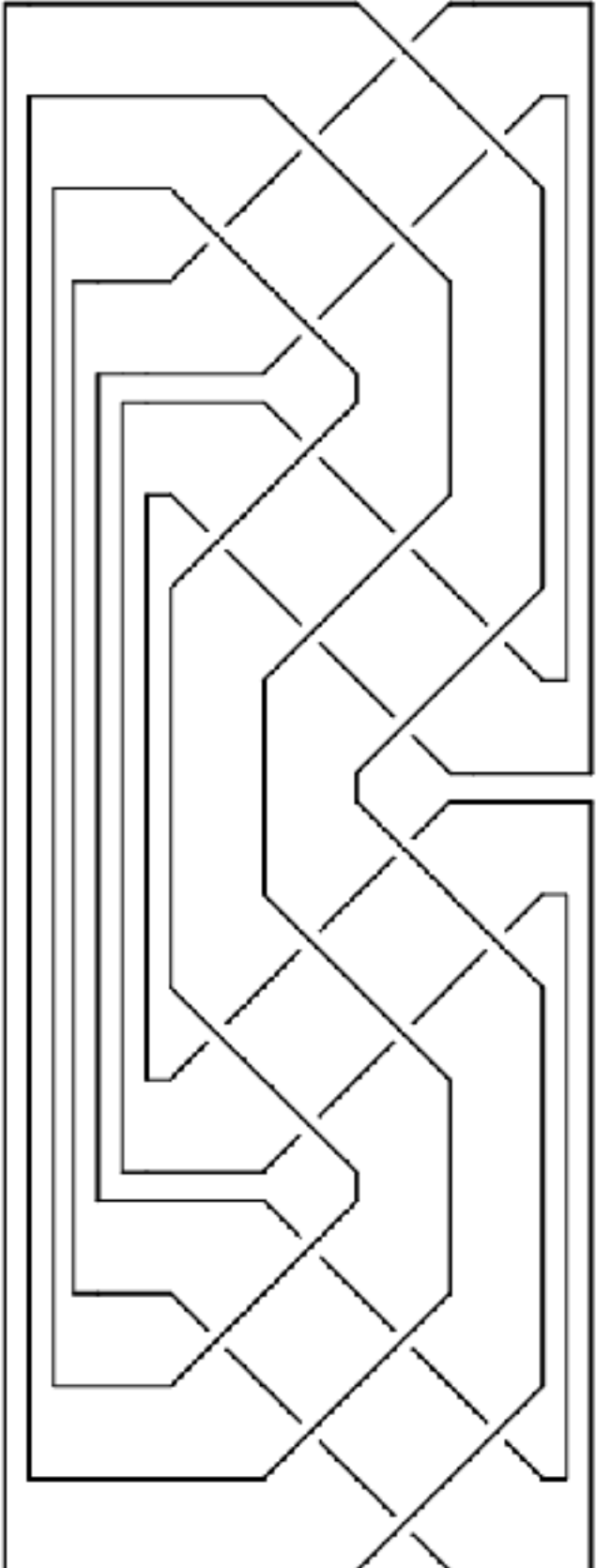
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Complete reducibility $\rightarrow$ trivial links

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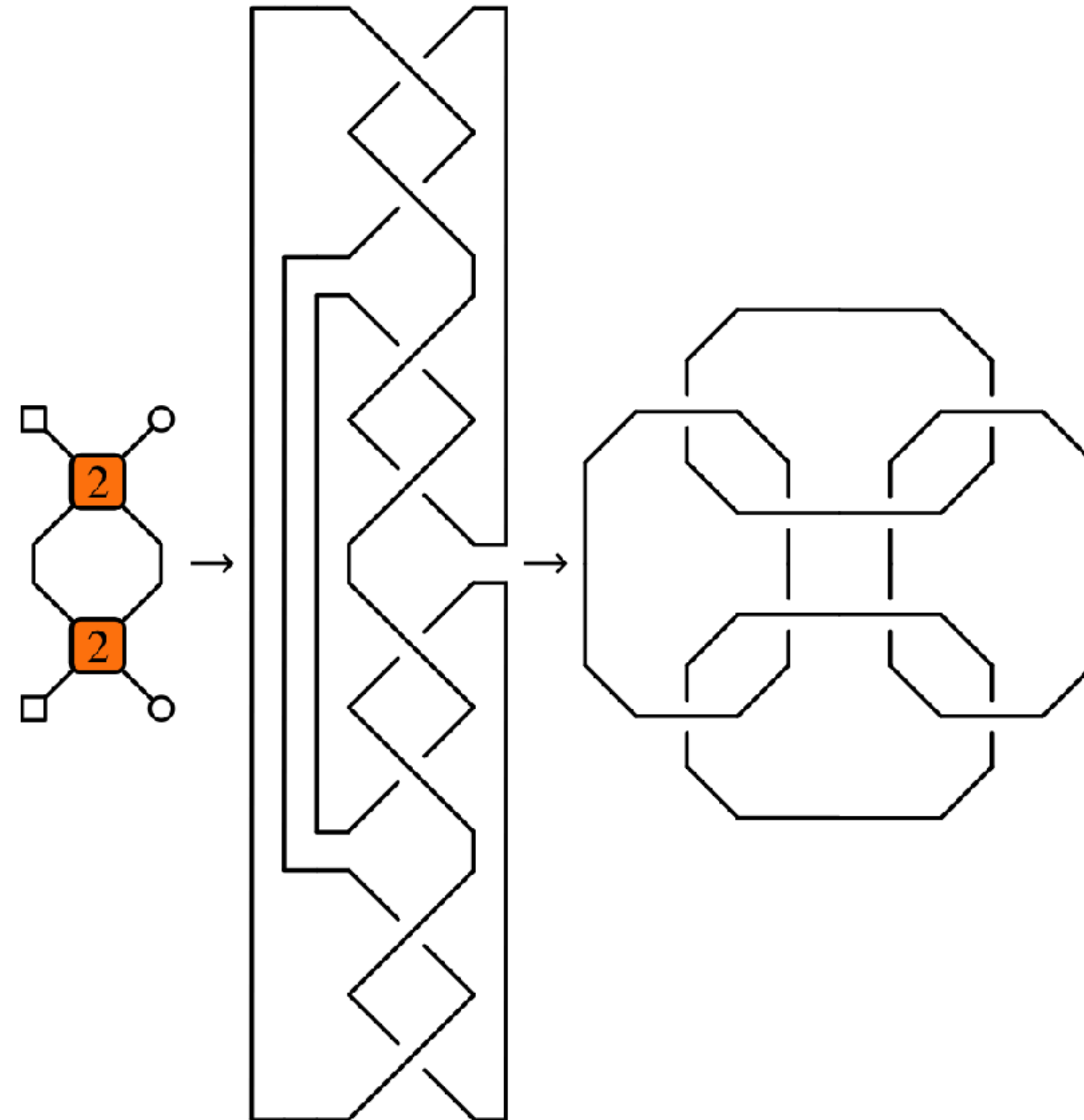
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$$Z_2(m, n) = \left(\frac{1}{d}\right)^{2(m+n)} \left[\text{Diagram 1} \right] = \left(\frac{1}{d}\right)^{2(m+n)} \left[\text{Diagram 2} \right] = \left(\frac{1}{d}\right)^{2(m+n) - N_\ell}$$



Example

- Non-reducible diagrams correspond to linked links



Summary

- Common framework for large class of solvable models
- New exactly solvable circuits
- Connection to knot theory

In collaboration with:



Suhail Rather, Freie Universität Berlin

Outlook and open questions

- Random geometries or measurements
- Analytical results on spectral correlations



Pieter Claeys, Trinity College Dublin

Backup Slides

Operator Dynamics



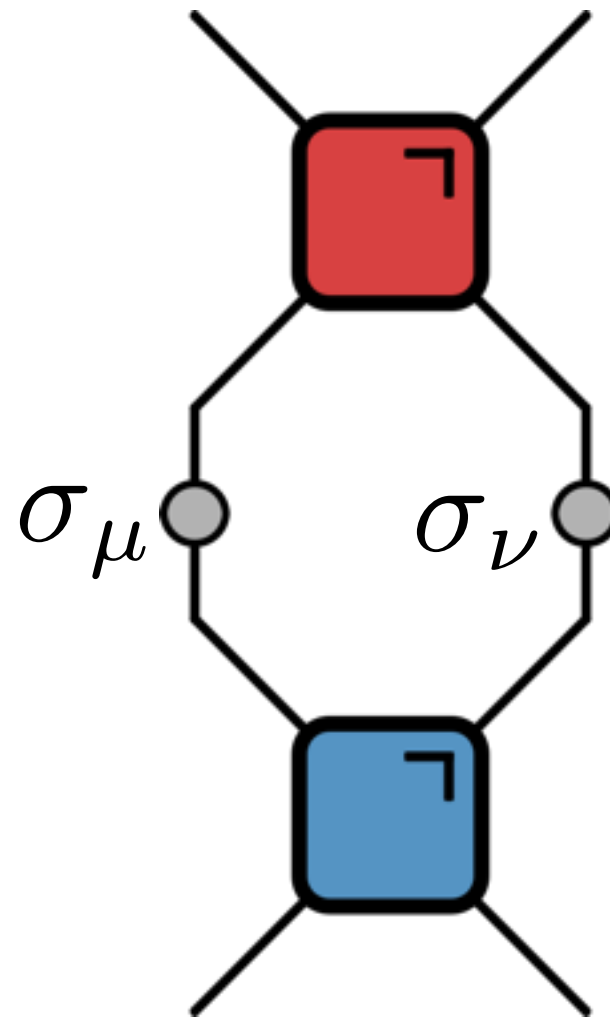
Operator evolution through super-operator

$$\sigma_{\mu} \otimes \sigma_{\nu} \rightarrow U^{\dagger}(\sigma_{\mu} \otimes \sigma_{\nu})U$$

Operator Dynamics

Operator evolution through super-operator

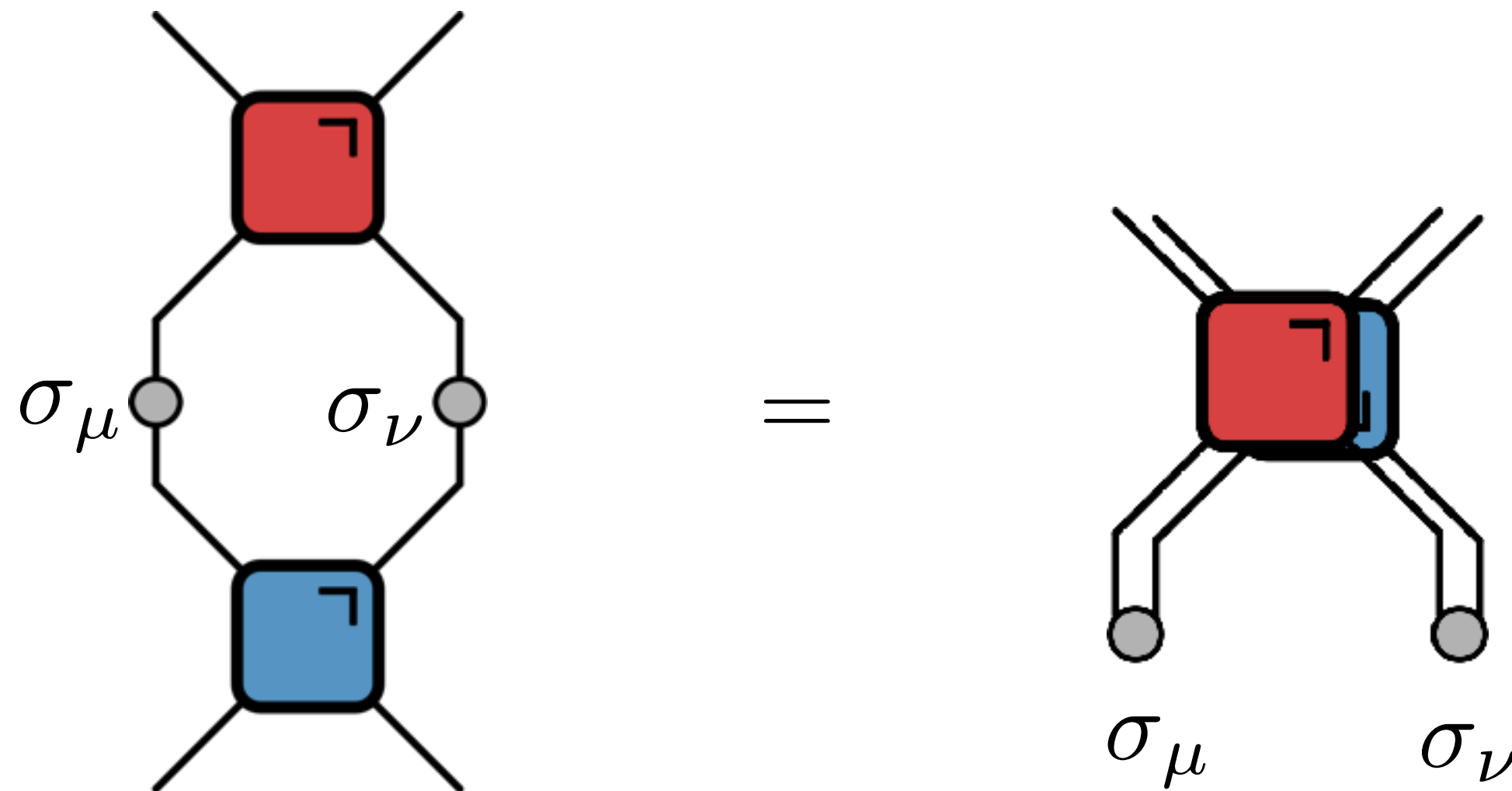
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Operator Dynamics

Operator evolution through super-operator

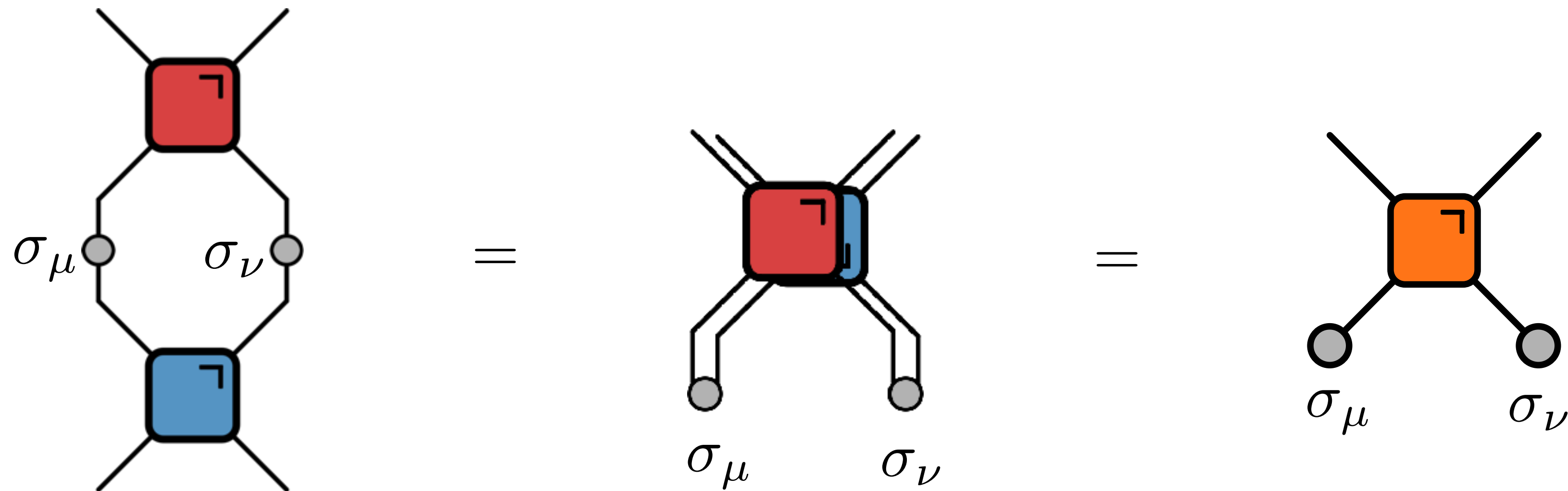
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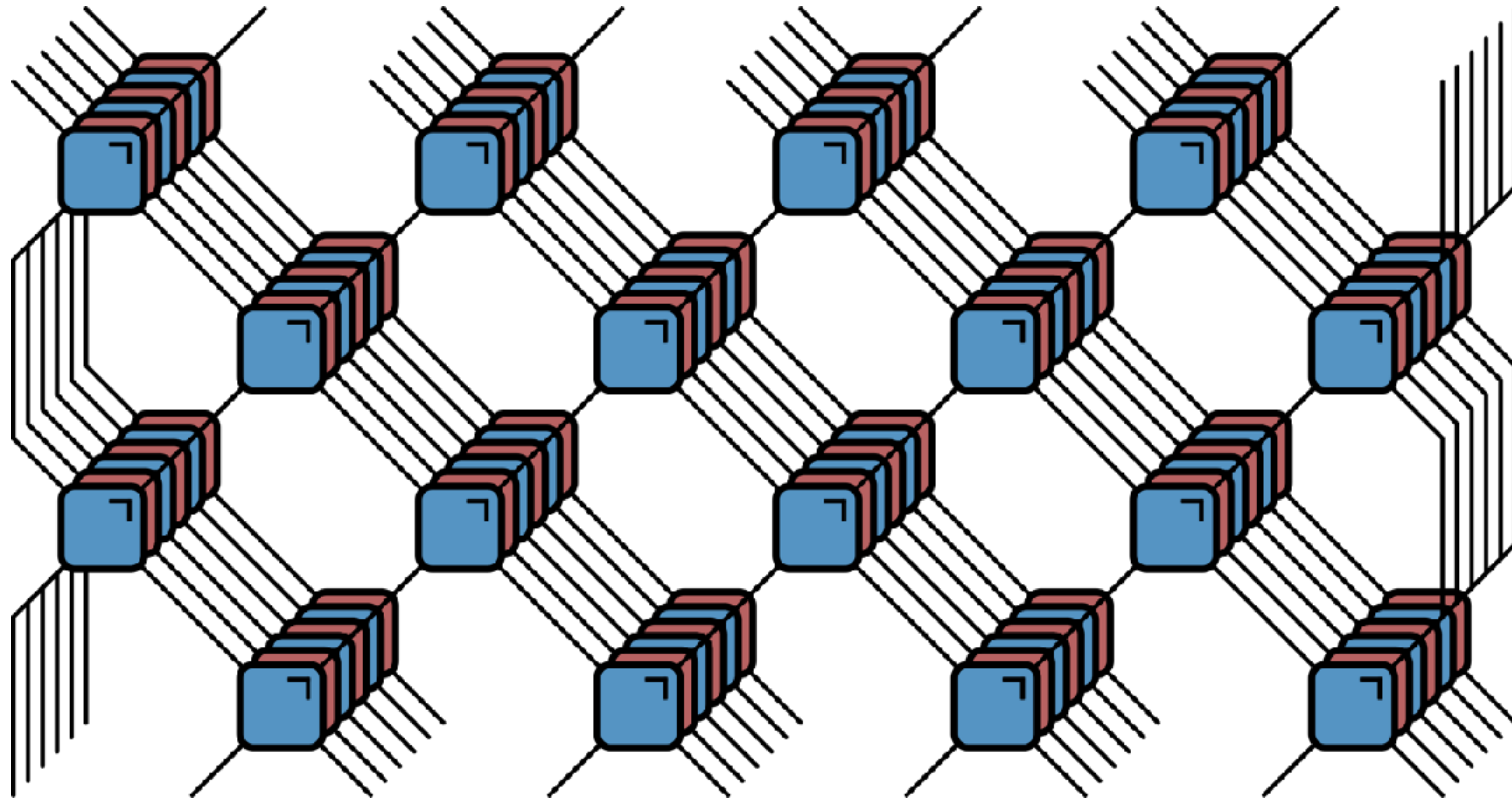
Operator Dynamics

Operator evolution through super-operator

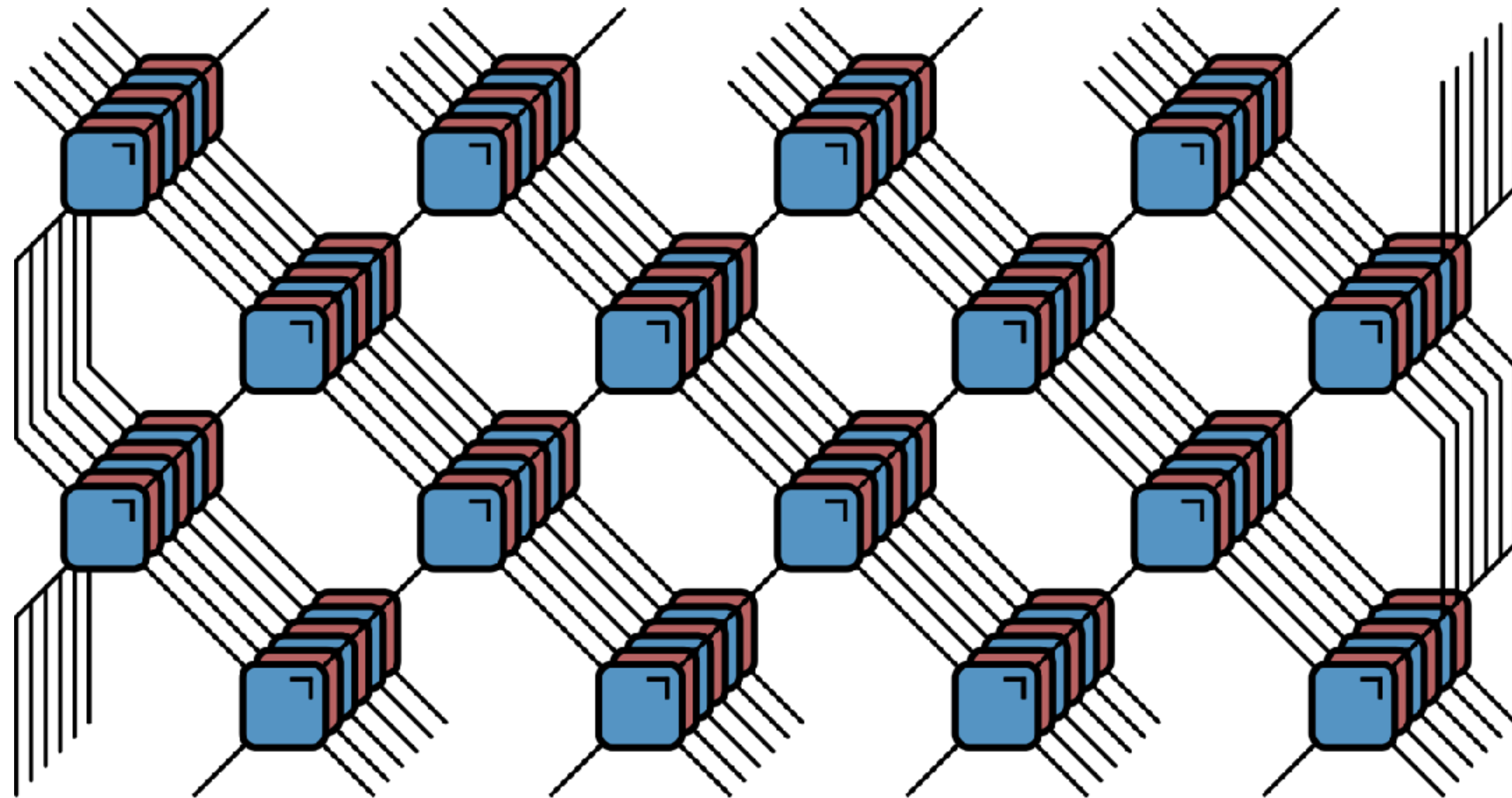
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Entanglement Membrane Theory

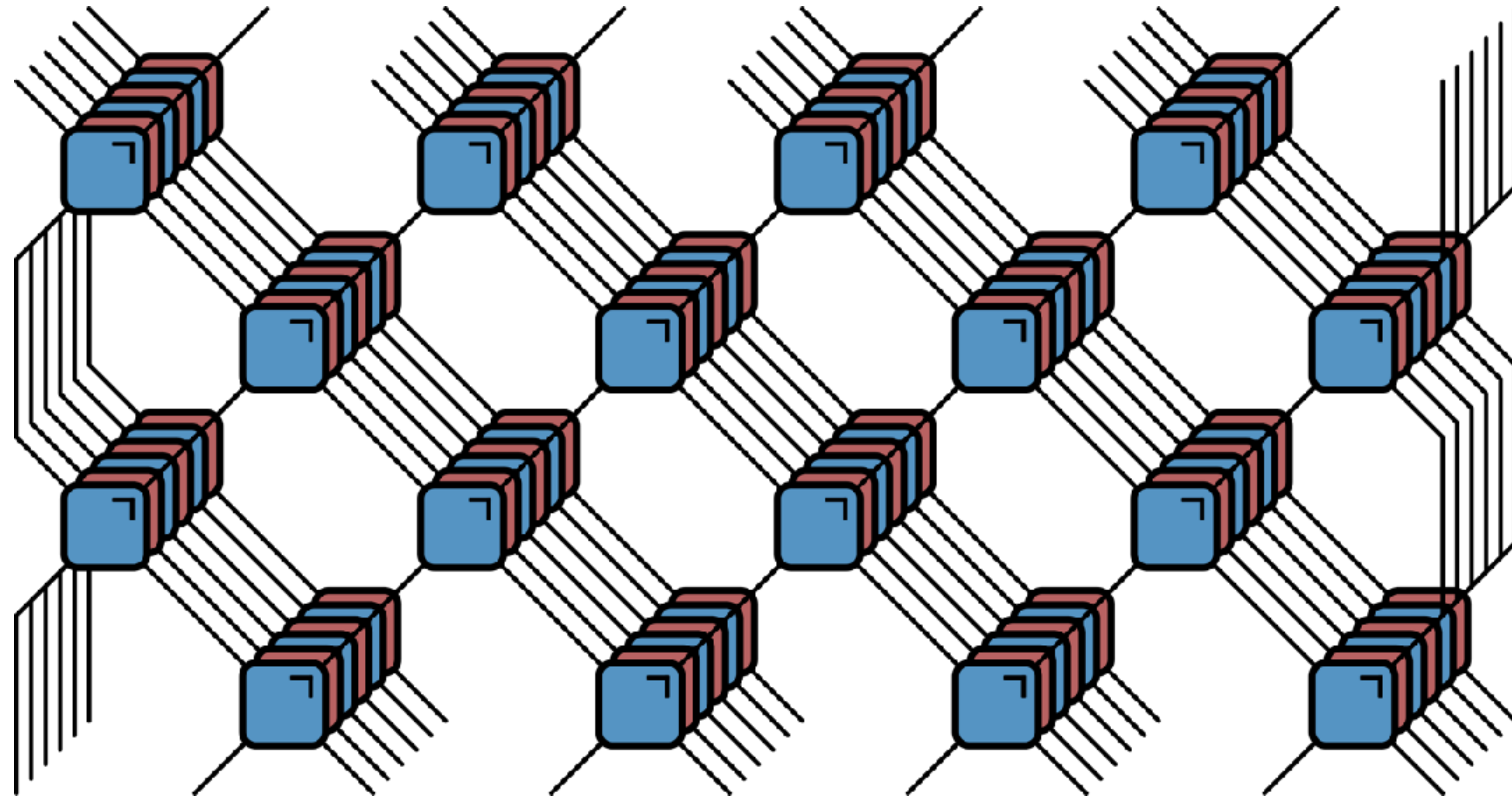


Entanglement Membrane Theory



- Phases on different Keldysh contours want to cancel $\rightarrow$ permutation-valued dof

Entanglement Membrane Theory



- Phases on different Keldysh contours want to cancel $\rightarrow$ permutation-valued dof
- Domain walls between different pairings of layers

Random Unitary circuits

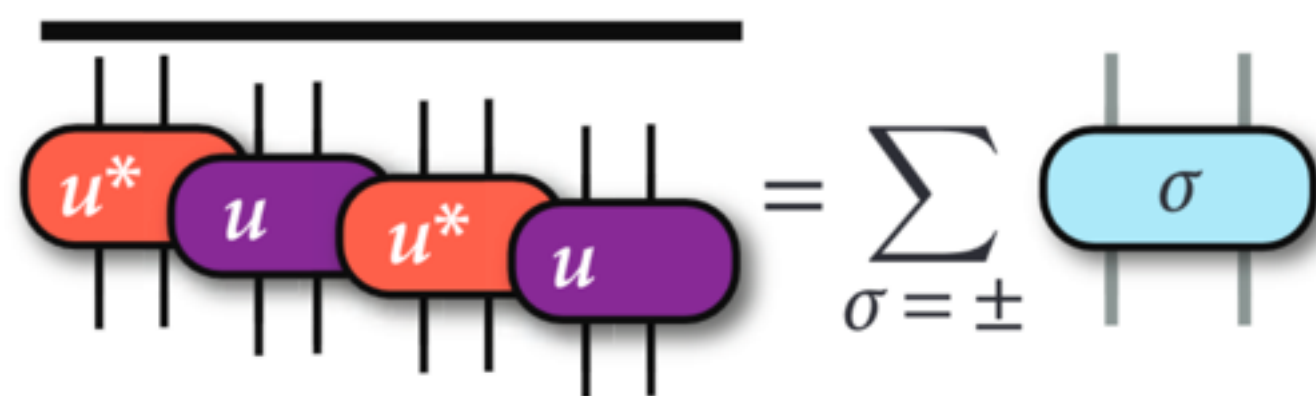
- Choose each gate independently from uniform distribution over $U(q^2)$

Dahlsten, Oliveira, Plenio
(2007), Nahum et al. (2018),
von Keyserlingk et al. (2018),
Khemani et al. (2018)

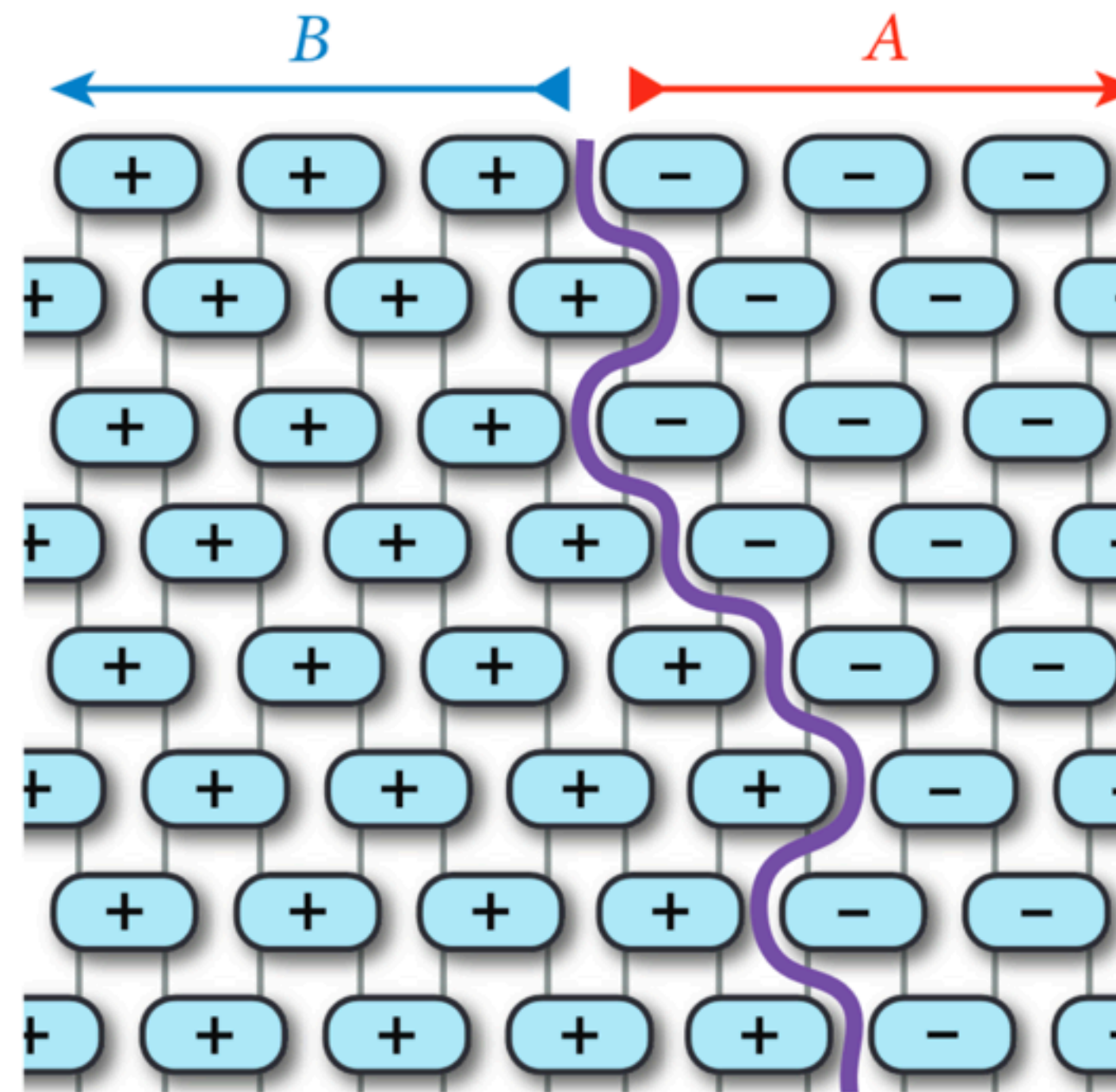
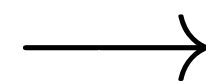
Random Unitary circuits

- Choose each gate independently from uniform distribution over $U(q^2)$
- Averaged tensor network is partition function of “magnet” with permutations as dof

Dahlsten, Oliveira, Plenio (2007), Nahum et al. (2018), von Keyserlingk et al. (2018), Khemani et al. (2018)



$$\overline{u^* u u^* u} = \sum_{\sigma = \pm} \sigma$$



Fisher et al. (2023)